



Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur 440033

**Scheme and Syllabus for
Bachelor of Technology (Electrical Engineering)**

**Submitted by
Board of Studies in Electrical Engineering**

FYUG Engineering Curriculum: NEP-Electrical Engineering

Vision

To provide quality professional education to aspiring students to produce globally competent technocrats, who can address challenges of industry and society to achieve sustainable socio-economic development.

Mission

- To produce graduates possessing sound fundamental knowledge of Electrical Engineering**
- To provide technical manpower for industry to solve problems with multidisciplinary approach.**
- To encourage students for scholarly research in emerging areas of electrical engineering**
- To inculcate ethical and social values.**

Program Outcomes (PO'S)

PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals ,and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSO)

PSO1	Interpret, identify, analyze, and evaluate problems in power system operation, control and design.
PSO2	Demonstrate knowledge to develop, control and assess electrical and electronic systems.

Program Educational Objectives (PEOs)

PEO1	Electrical engineer graduate shall be ready for modern technologies of electrical power system, energy industry and non-conventional energy sources.
PEO2	Electrical engineer graduate shall be able to enhanced analytical skill to solve industrial problem and work as a entrepreneur
PEO2	Electrical engineer graduate shall be able promote the awareness of green technologies by considering environmental aspects.

Semester wise Credit Distribution Structure for Four Year UG Electrical Engineering (Major) and Minor in Multidisciplinary

[illegible]

SCHEME OF TEACHING & EXAMINATION (I – VIII SEMESTER)

B.Tech.Sem –I (Electrical Engineering –Major)

SN	Course Category	Course Code	Name of Course	BoS	Teaching Scheme (hrs)			Total Credit	Examination Scheme						
									Theory				Practical		
					TH	TU	P		Exam Hrs	SEE	CIE	Min.	SEE	CIE	Min.
1	BSC - I	BEL1T01	Engineering Mathematics-I	ASH	3	-	-	3	3	70	30	45	-	-	-
2	BSC - II	BEL1T02	Applied Physics	ASH	2	-	-	2	3	70	30	45	-	-	-
3	BSC - II	BEL1P02	Applied Physics Lab	ASH	-	-	2	1	-	-	-	-	25	25	25
4	ESC – I	BEL1T03	Basics of Electrical & Electronics Engineering	EE	3	-	-	3	3	70	30	45	-	-	-
5	ESC – I	BEL1P03	Basics Electrical & Electronics Engineering Lab	EE	-	-	2	1	-	-	-	-	25	25	25
6	ESC – II	BEL1T04	Engineering Graphics	ME	3	-	-	3	3	70	30	45	-	-	-
7	ESC – II	BEL1P04	Engineering Graphics Lab	ME	-	-	2	1	-	-	-	-	-	50	25
8	AEC-I	BAE1T01	Communication Skills	ASH	1	0	0	1	2	35	15	23			
9	AEC-I	BAE1P01	Communication Skills	ASH	0	0	2	1	--	--	--	--	25	25	25
9a	VSC – I	BVS1P01	Basics of Solar PV Plant Installation	EE	-	-	4	2	-	-	-	-	50	50	50
10	CC-I	BCC1P01	Refer CC Basket	ASH	-	-	4	2	-	-	-	-	-	100	50
			Total		12	0	16	20		315	135		125	275	

B.Tech. Sem –II (Electrical Engineering –Major)

SN	Course Category	Course Code	Name of Course	BoS	Teaching Scheme (hrs)			Total Credit	Examination Scheme						
									Theory				Practical		
					TH	TU	P		Exam Hrs	SEE	CIE	Min.	SEE	CIE	Min.
1	BSC - III	BEL2T05	Engineering Mathematics-II	ASH	3	-	-	3	3	70	30	45	-	-	-
2	BSC - IV	BEL2T06	Applied Chemistry	ASH	3	-	-	3	3	70	30	45	-	-	-
3	BSC - IV	BEL2P06	Applied Chemistry Lab	ASH	-	-	2	1	-	-	-	-	-	50	25
4	ESC – III	BEL2T07	Engineering Mechanics	CV	3	-	-	3	3	70	30	45	-	-	-
5	ESC – III	BEL2P07	Engineering Mechanics Lab	CV	-	-	2	1	-	-	-	-	25	25	25
6	PCC – I	BEL2T08	Elements of Electrical Engineering & Measurements	EE	2	-	-	2	3	70	30	45	-	-	-
7	PCC – I	BEL2P08	Elements of Electrical Engineering & Measurements	EE	-	-	2	1	-	-	-	-	25	25	25
8	SEC – I	BSE2P01	Refer SEC Basket	-	-	-	4	2	-	-	-	-	50	50	50
9	IKS-I	BIK2T01	Refer IKS Basket	ASH	2	-	-	2	3	70	30	45	-	-	-
10	CC-II	BCC2P01	Refer CC Basket	ASH	-	-	4	2	-	-	-	-	-	100	50
			Total		13	0	14	20		350	150		100	250	

Exit option : Award of UG Certificate in Major with 40 credits and an additional 8 credits in skill based courses , internship , mini project etc

B.Tech. Sem –III (Electrical Engineering –Major)

SN	Course Category	Course Code	Name of Course	BoS	Teaching Scheme (hrs)			Total Credit	Examination Scheme						
									Theory				Practical		
					TH	TU	P		Exa m Hrs	SEE	CIE	Min.	SEE	CIE	Min.
1	PCC – II	BEL3T09	Network Analysis	EE	3	-	-	3	3	70	30	45	-	-	-
2	PCC – II	BEL3P09	Network Analysis Lab	EE	-	-	2	1	-	-	-	-	25	25	25
3	PCC – III	BEL3T10	Renewable Energy Sources	EE	3	-	-	3	3	70	30	45	-	-	-
4	PCC – III	BEL3P10	Renewable Energy Sources Lab	EE	-	-	2	1	-	-	-	-	25	25	25
5	MDM-I	BMD3T11	Electronic Devices & Circuits	EE	2	-	-	2	3	70	30	45	-	-	-
6	OE-I	BOE3T01	Open Elective-I		3	-	-	3	3	70	30	45	-	-	-
7	OE-I	BOE3P01	Open Elective-I Lab		-	-	2	1	-	-	-	-	25	25	25
8	HSSM	BHM3T01	Entrepreneurship & Startups	ASH	2	-	-	2	2	35	15	23	-	-	-
9	VEC	BVE3T01	Constitution of India	ASH	2	-	-	2	2	35	15	23	-	-	-
10	CEP	BCE3P01	Community Engg. Project	EE	-	-	4	2	-	-	-	-	-	100	50
			Total		15	0	10	20		350	150		75	175	

B.Tech. Sem –IV (Electrical Engineering –Major)

SN	Course Category	Course Code	Name of Course	BoS	Teaching Scheme (hrs)			Total Credit	Examination Scheme						
									Theory				Practical		
					TH	TU	P		Exam Hrs	SEE	CIE	Min.	SEE	CIE	Min.
1	PCC – IV	BEL4T11	Electrical Machines-I	EE	3	-	-	3	3	70	30	45	-	-	-
2	PCC – IV	BEL4P11	Electrical Machines-I Lab	EE	-	-	2	1	-	-	-	-	25	25	25
3	PCC – V	BEL4T12	Power Systems-I	EE	3	-	-	3	3	70	30	45	-	-	-
4	PCC – V	BEL4P12	Power Systems-I Lab	EE	-	-	2	1	-	-	-	-	25	25	25
5	MDM-II	BMD4T13	Elements of Electromagnetic	EE	2	-	-	2	3	70	30	45	-	-	-
6	OE-II	BOE4T02	Open Elective-II		2	-	-	2	3	70	30	45	-	-	-
7	AEC-II	BAE4P02	Technical Report Writing	EE	-	-	4	2	-	-	-	-	50	50	50
8	HSSM-II	BHM4T02	Digital Economy	ASH	2	-	-	2	2	35	15	23	-	-	-
9	VEC-II	BVE4T02	Universal Human Values	ASH	2	-	-	2	2	35	15	23	-	-	-
10	VSC-II	BVE4P02	Electrical Engineering Drawing	EE	-	-	4	2	-	-	-	-	50	50	50
			Total		14	0	12	20		350	150		150	150	

Exit option : Award of UG Diploma in Major with 80 credits and an additional 8 credits in skill based courses , internship , mini project etc

B.Tech. Sem –V (Electrical Engineering –Major)

SN	Course Category	Course Code	Name of Course	BoS	Teaching Scheme (hrs)			Total Credit	Examination Scheme						
									Theory				Practical		
					TH	TU	P		Exam Hrs	SEE	CIE	Min.	SEE	CIE	Min.
1	PCC – VI	BEL5T13	Electrical Machines-II	EE	3	-	-	3	3	70	30	45	-	-	-
2	PCC – VI	BEL5P13	Electrical Machines-II Lab	EE	-	-	2	1	-	-	-	-	25	25	25
3	PCC – VII	BEL5T14	Electrical Power Utilization	EE	2	-	-	2	3	70	30	45	-	-	-
4	PCC – VII	BEL5P14	Electrical Workshop Lab.	EE	-	-	2	1	-	-	-	-	50	50	50
5	PCC – VIII	BEL5T15	Power Systems-II	EE	3	-	-	3	3	70	30	45	-	-	-
6	PEC-I	BEL5T16	Program Elective-I (Refer Program Elective Basket)	EE	3	1	-	4	3	70	30	45	-	-	-
7	MDM-III	BMD5T17	Programming Techniques & Simulation	EE	3	-	-	3	3	70	30	45	-	-	-
8	MDM-III	BMD5P17	Programming Techniques & Simulation Lab	EE	-	-	2	1	-	-	-	-	25	25	25
9	OE-III	BOE5T03	Open Elective-III		2	-	-	2	3	70	30	45	-	-	-
			Total		16	1	6	20		420	180		100	100	

B.Tech. Sem –VI (Electrical Engineering –Major)

SN	Course Category	Course Code	Name of Course	BoS	Teaching Scheme (hrs)			Total Credit	Examination Scheme						
									Theory				Practical		
					TH	TU	P		Exam Hrs	SEE	CIE	Min.	SEE	CIE	Min.
1	PCC – IX	BEL6T18	Power Electronics	EE	3	-	-	3	3	70	30	45	-	-	-
2	PCC – IX	BEL6P18	Power Electronics Lab	EE	-	-	2	1	-	-	-	-	25	25	25
3	PCC – X	BEL6T19	Control System	EE	3	1	-	4	3	70	30	45	-	-	-
4	PCC – XI	BEL6P19	Control System Lab	EE	-	-	2	1	-	-	-	-	25	25	25
5	PEC-II	BEL6T20	Program Elective-II	EE	3	-	-	3	3	70	30	45	-	-	-
6	PEC-III	BEL6T21	Program Elective-III	EE	3	1	-	4	3	70	30	45	-	-	-
7	MDM-IV	BMD6T22	Microprocessor & Microcontroller	EE	2	-	-	2	3	70	30	45	-	-	-
8	SEC-II	BSE6P02	Electrical Installation Design lab	EE	-	-	4	2	-	-	-	-	50	50	50
			Total		14	2	8	20		350	150		100	100	

Exit option : Award of UG Degree B.Voc / B.Sc. in Major with 120 credits and an additional 8 credits in skill based courses , internship , mini project etc.

B.Tech. Sem –VII (Electrical Engineering –Major)

SN	Course Category	Course Code	Name of Course	BoS	Teaching Scheme (hrs)			Total Credit	Examination Scheme						
									Theory				Practical		
					TH	TU	P		Exam Hrs	SEE	CIE	Min.	SEE	CIE	Min.
1	RM	BRM7T23	Research Methodology #	EE	3	-	-	3	3	70	30	45	-	-	-
2	RM	BRM7P23	Effective writing#	EE	1	-	-	1	-	35	15	23	-	-	-
3	PEC-IV	BEL7T24	Program Elective-IV #	EE	2	-	-	2	3	70	30	45	-	-	-
4	MDM-V	BMD7T25	Introduction to Artificial Intelligence & Machine Learning #	EE	2	-	-	2	3	70	30	45	-	-	-
5	OJT	BOJ7P01	Internship (12 Weeks)	EE	-	-	24	12	-	-	-	-	200	200	200
			Total		8	0	24	20		245	105		200	200	

Indicates that, Online Courses to be done from NPTEL. Examination will be conducted by NPTEL / RTMNU

B.Tech. Sem –VIII (Electrical Engineering –Major)

SN	Course Category	Course Code	Name of Course	BoS	Teaching Scheme (hrs)			Total Credit	Examination Scheme						
									Theory				Practical		
					TH	TU	P		Exam Hrs	SEE	CIE	Min.	SEE	CIE	Min.
1	PCC – XII	BEL8T26	Power System Protection	EE	3	-	-	3	3	70	30	45	-	-	-
2	PCC – XII	BEL8P26	Power System Protection Lab	EE	-	-	2	1	-	-	-	-	25	25	25
3	PCC – XIII	BEL8T27	Computer Application in Power Systems	EE	3	-	-	3	3	70	30	45	-	-	-
4	PCC – XIII	BEL8P27	Computer Application in Power Systems Lab	EE	-	-	2	1	-	-	-	-	-	50	25
5	PEC-V	BEL8T28	Program Elective-V	EE	3	-	-	3	3	70	30	45	-	-	-
6	PEC-VI	BEL8T29	Program Elective-VI	EE	3	-	-	3	3	70	30	45	-	-	-
7	MDM-VI	BMD8T30	IoT Application in Electrical Engg.	EE	2	-	-	2	3	70	30	45	-	-	-
8	Project	BPR8P01	Project	EE	-	-	8	4	-	-	-	-	100	100	100
			Total		14	0	12	20		350	150		125	175	

4-Years Bachelor's Degree (B.Tech.) in Electrical Engineering with Multidisciplinary Minor

Abbreviations

OE	Generic/ Open Electives
VSEC	Vocational Skill and Skill Enhancement Courses
VSC	Vocational Skill Courses
SEC	Skill Enhancement Courses
AEC	Ability Enhancement Courses
IKS	Indian Knowledge System
VEC	Value Education Courses
OJT	On Job Training: Internship/ Apprenticeship.
FP	Field projects
CEP	Community engagement project
CC	Co-curricular Courses
RM	Research Methodology
RP	Research Project
Lib. Learn	Liberal Learning Course
HSSM	Courses on Humanities, Social Science, and Management
SEE	Semester End Examination
CIE	Continuous Internal Evaluation

LIST OF PROGRAM ELECTIVE COURSES (PEC) ELECTRICAL ENGINEERING

S.N.	Semester	Category	Course code	Course name
1	5 th	PEC-I	BEL5T16A	PLC and SCADA
2			BEL5T16B	Electrical Machine Design
3			BEL5T16C	Digital Electronics circuits
4	6 th	PEC-II	BEL6T20A	Electrical Safety & Standards
5			BEL6T20B	Electric Drives & Control
6			BEL6T20C	Energy Management & Audit
7		PEC-III	BEL6T21A	Electric Vehicles
8			BEL6T21B	HV Engineering
9			BEL6T21C	Electrical Distribution Systems
10	7 th	PEC-IV	BEL7T24A#	EHVAC & HVDC Transmission
11			BEL7T24B#	Advance Control System
12			BEL7T24C#	Power Semiconductor Based Drives
13	8 th	PEC-V	BEL8T28A	Power Quality
14			BEL8T28B	Smart Grid System
15			BEL8T28C	Power Station Practice
16		PEC-VI	BEL8T29A	Flexible AC Transmission Systems
17			BEL8T29B	Solar Photovoltaic Systems
18			BEL8T29C	Battery Management System.

#indicates that Online Courses to be done from NPTEL. Examination will be conducted by NPTEL/RTMNU.

LIST OF OPEN ELECTIVES (OE) OFFERED BY ELECTRICAL ENGINEERING BOS

(Students of Electrical Engineering will not be able to opt these OE courses)

S.N.	Semester	Course code	Course name
1	3 rd SEM	BOE3T01	Fundamental of Electrical Engineering
2	4 th SEM	BOE4T02	Introduction to Power Systems
3	5 th SEM	BOE5T03	Power Plant Engineering

LIST OF INDIAN KNOWLEDGE SYSTEM COURSES (IKS) OFFERED BY ELECTRICAL ENGINEERING BOS

(Offered by Applied Science Board and Humanities)

S.N.	Semester	Category	Course code	Course name
1	2 nd sem	IKS-I	BIK2T01A	Consciousness Studies
2			BIK2T01B	Preserving Art, Culture and Tradition
3			BIK2T01C	Wellness, traditional medicines and yoga
4			BIK2T01D	Glimpses of ancient Science and Technology

LIST OF SKILL ENHANCEMENT COURSES (SEC) OFFERED BY ELECTRICAL ENGINEERING BOS

S.N.	Semester	Course Code	Course Name
1	2 nd SEM	BSE2P01	Electric Wiring & Illumination System
2	6 th SEM	BSE6P02	Electrical Installation Design

First Semester Syllabus

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: I	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL1TO1	Engineering Mathematics -1	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives

1	The topics covered will equip them the techniques to understand advanced level mathematics and its applications that would enhance analytical thinking power.
2	The aim is to inculcate and develop the basic mathematic skills of engineering students that are imperative for effective understanding of engineering subjects.

Course Outcomes

After completion of syllabus, students would be able to

1	Analyse real world scenarios to recognize when derivatives or integrals are appropriate, formulate problems about the scenarios, creatively model these scenarios (using technology, if appropriate) in order to solve the problems using multiple approaches, judge if the results are reasonable, and then interpret and clearly communicate the results.
2	Appreciate ODE and system of ODEs concepts that are encountered in the real world, understand and be able to communicate the underlying mathematics involved to help another person gain insight into the situation.
3	Apply knowledge of mathematics, physics and modern computing tools to scientific and engineering problems.
4	Develop an ability to identify, formulate and/or solve real world problems.
5	Understand the impact of scientific and engineering solutions in a global and societal context.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1 : Differential Calculus			
Successive differentiation: Leibnitz's Rule, Taylor's and Maclaurin's series for function of one variable, Indeterminate forms and L'Hospital's Rule.	5		1
Unit 2: Multivariable Calculus (Differentiation)			
Functions of several variables, First and Higher order partial derivatives, Euler's theorem, Chain rule and Total differential coefficient, Jacobians, Lagrange's method of undetermined multipliers.	5		2
Unit 3: Matrices			
Rank of a matrix, Consistency of linear system of non-homogeneous equations, Linear dependence of vectors, Eigen values and Eigen vectors, Reduction to diagonal form, Cayley-Hamilton theorem.	4		3
Unit 4: First Order Ordinary Differential Equations			
Linear, Reducible to linear and Bernoulli's differential equations, Exact differential equations (Excluding the cases of integrating factors), Application of first order differential equation to simple electrical circuits.	5		4
Unit-5: Higher Order Ordinary Differential Equations			
Higher order ordinary linear differential equations with constant coefficients, Method of variation of parameters, Cauchy's and Legendre's homogeneous differential equations, Applications of higher order differential equations to simple electrical circuits.	5		5

Text/Reference Books:

- (1) Erwin Krayzig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
- (2) Ramana B.V., Higher Engineering Mathematics, Tata Mc-Graw Hill, New Delhi, 11th Reprint, 2010.
- (3) N.P. Bali and Manish Goyal, A textbook of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
- (4) B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
- (5) P. N. Wartikar and J. N. Wartikar, Applied Mathematics, Volume I and II.
- (6) H.K Dass, Rama Verma, Rajnish Verma, V.J. Dagwal, Sajid Anwar and D.F. Shastrakar, Engineering Mathematics, Volume I and II, S. Chand.

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B.TECH. ELECTRICAL ENGINEERING**

Sem: I	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL1TO2	Applied Physics	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Learn the basic concepts of the dual nature of matter and wave packet and apply them to analyze various relevant phenomena and to solve related numerical problems.
2	Apply concepts in interference and diffraction to solve relevant numerical problems and to relate to relevant engineering applications
3	Identify and explain different types of diodes, transistors, and its applications.
4	Learn and explain quantum transitions and apply them to the working of lasers.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1 :Quantum Mechanics			
Concept of wave-particle duality, De-Broglie Hypothesis, Matter Waves, Davisson-Germer Experiment; Bohr's Quantization condition. Concept of wave packets, Heisenberg Uncertainty Principle. Schrodinger wave equation (time dependent and time independent), Wave function Ψ and normalization condition, Application to one dimensional infinite potential well.	6		1
Unit 2: Wave optics			

Huygen's principle, superposition of waves and interference of light by wavefront splitting and amplitude splitting, Interference in thin films, Interference in Wedge-shaped thin film, Newton's rings, and Anti-reflection coating.	6		2
Unit 3: Semiconductor Devices			
Semiconductor, Classification, Pure and doped semiconductors. Types of Semiconductor diodes -P-N junction Diode, Tunnel Diode, Zener Diode, Light Emitting Diode (LED), Photodiode. Transistors, Hall Effect, Hall voltage, and Hall coefficient; its applications	6		3
Unit 4: Lasers			
Quantum Transitions: Absorption, Spontaneous emission & stimulated Emission, Metastable states, Principle of laser, Laser characteristics, Coherence length and coherence time, Pumping schemes: Three level and Four level. Optical Resonator, Construction & working of Ruby laser and He-Ne laser, Semiconductor diode laser, Applications of laser.	6		4

Suggested Text Books & Reference Books

1. P. M. Mathews and K. Venkatesan, A Textbook of Quantum Mechanics, Tata Mc Graw Hill (1977).
2. J. L. Powell and B. Crasemann, Quantum Mechanics, Narosa Publishing House (1993).
3. Charles Kittel, Introduction to Solid State Physics, Wiley Eastern, 5th edition, (1983).
4. A. J. Dekker, Solid State Physics, Prentice Hall of India (1971).
5. A Textbook of Engineering Physics, Dr. M. N. Avdhanulu, Dr. P. G. Kshirsagar, S. Chand Publication
6. Text book of Applied Physics, Dr. D. S. Hardas, Dr. D. S. Bhoomik, Dr. S. Shastri, Das Ganu Publication ISBN-978-93-84336-59-2 (2021)
7. Applied Physics, M. N. Avdhanulu, Shilpa A. Pande, Arti R. Golhar, Mohan Giriya, S. CHAND
8. A Text Book of Engineering Physics Dr. Devashree Hardas & Dr. Ashish Panat, Das Ganu Publication ISBN-978-81-921757-7-5 (2011)
9. Applied Physics, - Dr. (Mrs) S.P. Wankhede, Dr. Shruti Patle, Dr. (Mrs.) S.U. Bhonsule and Dr. N. S. Ugemuge DNA Publication ISBN-978-81-945174-6-7 (2020)
10. Quantum Physics of Atoms, Molecules, Solids, Nuclei, and Particles by R. Eisberg and R. Resnick, Wiley and Sons
11. Engineering Physics, second edition, Sanjay Jain, G. Sahasrabudhe, University's Press (India) Pvt. Ltd. (2016)
12. D. J. Griffiths, Quantum mechanics, Prentice Hall of India Private Limited, New Delhi
13. L. I. Schiff, Quantum Mechanics, TMH Publications
14. Advanced Engineering Materials - Dr. Sangeeta G. Itankar, Dr. Manjusha Dandekar, Dr. Tushar R. Shelke, Dr. Swati Fartode, Alliance & Co. ISBN 978-93-91322-12-0 (2023)
15. Applied Physics- Dr. Sangeeta G. Itankar, Dr. Manjusha Dandekar, Dr. Tushar R. Shelke, Dr. Swati Fartode, Alliance & Co. ISBN 978-93-91322-97-7 (2023)
16. David Halliday, Robert Resnick, Jearl Walker, Principles of Physics, 10th Edition, John Wiley and Sons (2017)
17. Advanced physics - Dr. Shruti Patle, Dr. (Mrs.) S.U. Bhonsule, Dr. Ashish N. Bodhaye, Dr. Manohar D. Mehare DNA Publication (2019)
18. Engineering Physics - Dr. N. S. Ugemuge, Dr. (Mrs.) S.U. Bhonsule and Dr. Shruti Patle DNA Publication (2019)

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FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: I	Total Hours Distribution per week		
Total Credit :1	Practical (P): 2 Hrs.		
Subject Code	BEL1P02	Applied Physics Lab.	
Examination Scheme - Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	--

LIST OF PRACTICAL:

Pr. No.	List of Practical
1	Energy gap of semiconductor /thermistor.
2	Parameter extraction from V-I characteristics of PN junction diode.
3	Parameter extraction from V-I characteristics of Zener diode.
4	Parameter extraction from V-I characteristics of PNP/NPN transistor in CB and CE mode.
5	V-I Characteristics of Tunnel diode.
6	V-I Characteristics of Light Emitting Diodes/ Determination of Plank's constant by using LEDs.
7	Study of Diode rectification.
8	Study of Hall Effect and determination of Hall Voltage of a given sample.
9	Variation of Hall coefficient (RH) with temperature.
10	To study B-H curve and to find out the values of coercivity, retentivity, and saturation magnetization of the experimental material.
11	Determination of NA for optical fiber
12	Calibration of Time Base circuit of CRO and determination of AC, DC voltage & frequency of electrical signals using CRO.
13	To determine the number of lines per cm on a diffraction grating using a LASER beam.
14	Virtual Lab: Experiment on the determination of the thickness of a thin foil using an air wedge arrangement.
15	Virtual Lab: Experiment on the determination of the refractive indices of the material corresponding to ordinary and extra - ordinary rays.

Note: Performance of at least six experiments is compulsory in a semester.

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Sem: I	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL1TO3	Basics of Electrical & Electronics Engineering	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives	
1	To learn basic concepts and principles of Electrical and Electronics Engineering.
2	To Understand basic operation and working of electrical machines.
3	To learn the concepts of electrical & electronic circuits .

Course Outcomes	
After completion of syllabus, students would be able to	
1	Understand basic properties of electrical components and electrical parameters
2	Analyse DC Electric & Magnetic Circuits
3	Learn and analyse AC circuits
4	Understand working principle of electrical machines
5	Know the electronic devices and their properties & use the various electronic devices for various applications.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1			
Circuit Elements and Parameters- Resistance, Inductance & Capacitance, Basic relations between voltage, current and resistance, classification of material according to conductivity, various effects current.	7		1

Unit 2:			
Electrical Circuit Fundamentals -DC Circuits: Types, Series, Parallel, Star, Delta and transformation Kirchhoff's Laws & Applications, Energy Sources and Circuit Reduction, Magnetic Circuits: Flux, MMF, Reluctance, Analogy with Electric Circuits. Simple Calculations for Composite Magnetic Circuits	5		2
Unit 3:			
AC Circuits & Fundamentals -Principle of Generation of Single Phase , Periodic Function, Average & R.M.S., Values, Steady State Behavior With Sinusoidal Excitation, Phasor Representation, Reactance & Impedance, Series & Parallel Circuit, Power Factor.	10		3
Unit 4:			
Introduction to Electrical Machines - Transformer: Basic Principles, Construction, Phasor Diagram for Transformer under no load and on load conditions, Voltage Regulation and Efficiency, DC Machines: Introduction and types, Working Principle, simple mathematical equations, Other types of machines: Induction Motor, Stepper Motors (Only Working Principle)	8		4
Unit-5:			
Introduction to Electronic Components - P-N Junction Diode, Bi-junction transistor, UJT, FET etc and their properties, Diode as a rectifier. Fundamental Applications of Transistors - Common emitter, common collector and common base configurations of BJT, Emitter follower, various Biasing circuits for BJT, BJT as a switch, BJT as an amplifier. Basic introduction to OP-AMP	9		5

Text & Reference Books:

1. D. C. Kulshrehtha, "Basic Electrical Engineering", Tata Mcgraw Hill, 2012
2. B.L.Theraja, "Electrical Technology", S.Chand
3. MillmanHalkias , "Electronic Devices and Circuits", Tata McGraw Hill, 2000

Online references:

1. Basic Electrical Engineering A Web course of NPTEL by Day, Bhattacharya & Roy, Available:- [www. nptel.ac.in](http://www.nptel.ac.in)

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Sem: I	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL1TO4	Engineering Graphics	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives

1	To improve imagination skills.
2	Learn to sketch and take field dimensions.
3	Learn to take data and transform it into graphic drawings.
4	Learn basic engineering drawing formats.

Course Outcomes

After completion of syllabus, students would be able to

1	Get acquainted with the knowledge of various lines, geometrical constructions and construction of various kinds of scales, and Ellipse.
2	Improve their imagination skills by gaining knowledge about points, lines and planes.
3	Become proficient in drawing the projections of various solids.
4	Gain knowledge about orthographic and isometric projections.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1			
Polygons-Construction of Regular Polygons using given length of a side; Ellipse-General method and Oblong Methods for Construction of ellipse ;Scales-Plain, Vernier and Diagonal Scales. Introduction to Orthographic Projections; Projections of Points; Projections of Straight Lines parallel to both planes; Projections of Straight Lines-Parallel to	7		1

one and inclined to other plane.			
Unit 2:			
Projections of Straight Lines inclined to both planes, determination of true lengths, angle of inclinations and traces.	5		2
Unit 3:			
Projections of Planes; Regular Planes Perpendicular / Parallel to one Reference Plane and inclined to other Reference Plane; Inclined to both the Reference Planes.	10		3
Unit 4:			
Projections of Solids-Prisms, Pyramids, Cylinders and Cones with the axis inclined to one Plane.	8		4
Unit-5:			
Conversion of Isometric Views to Orthographic Views. Conversion of Orthographic Views to Isometric Projections and Views.	9		5

Text Book:

1.Engineering Drawing by N.D.Bhat, Chariot publications

Reference Books:

1. Engineering Drawing by M.B. Shah and B.C. Rana, Pearson publishers
2. Engineering Drawing by Dhananjay A. Jolhe, Tata McGraw Hill Publishers
3. Engineering Graphics for Degree by K.C. John, PHI Publishers

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Sem: I	Total Hours Distribution per week		
Total Credit :1	Practical (P): 2 Hrs.		
Subject Code	BEL1P04	Engineering Graphics Lab.	
Examination Scheme - Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
50 Marks	--	25 Marks	--

LIST OF PRACTICAL: Based on the Theory Subject Syllabus as BEL1TO4

Based on the above syllabus (Project assign /Preparation of sheets)

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Sem: I	Total Hours Distribution per week		
Total Credit :1	Practical (P): 2 Hrs.		
Subject Code	BEL1P01	Basics of Electrical & Electronics Engineering Lab.	
Examination Scheme - Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	--

Practical Course Objectives	
1	To learn basics of Electrical and Magnetic Circuits
2	To learn the basic semiconductor Electronic devices and their applications

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Understand basic properties of electrical components and electrical parameters
2	Analyze DC Electric & Magnetic Circuits
3	Learn and analyze AC circuits
4	Know the electronic devices and their properties
5	Use the various electronic devices for various applications

LIST OF PRACTICAL:

Pr. No.	List of Practical
1	To study and verification of Kirchhoff's Laws applied to direct current circuit
2	To obtain the B/H curve of a magnetic material
3	To study AC series circuits
4	To Study AC parallel circuits
5	To study characteristics of P-N Junction diode
6	To study Half and Full wave rectifier
7	To study Transistor Characteristics
8	To study of BJT as amplifier

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Sem: I	Total Hours Distribution per week		
Total Credit : 1	Lecture (L): 1Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BAE1TO1	Communication Skills	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
15 Marks	35 Marks	23 Marks	2 Hours

Prerequisites: Basic knowledge of Communication Skills

Course Objectives	
1	Students would be able to enhance their communication skills.

Course Outcomes	
After completion of syllabus, students would be able to	
1	Construct grammatically correct sentences.
2	Identify and overcome barriers of communication.
3	Demonstrate good Listening and speaking skills.
4	Develop effective reading and writing skills.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1			
Grammar: Tenses and its types, sentences and its Types, Transformation of Sentences(Assertive, Affirmative, Negative, Interrogative, Exclamatory) Reported speech.	4		1
Unit-2			
Introduction to Communication, Importance of communication Types of communication -Verbal and non-verbal Communications: - Kinesics, Vocalics, Chronemics, Haptics, Proxemics), Barriers to communication and methods to overcome them.	3		2
Unit-3			

Introduction to LSRW Skills-, Listening Skills: Importance of listening, Types of listening, listening barriers and methods to overcome, Speaking Skills: Components of public speaking, Essential steps for public speaking, Overcoming stage fear in public speaking, Do's, and Don'ts of Public speaking	5		3
Unit-4			
Reading Skills: Importance of reading skills, Types of reading, comprehending passages, Writing Skills: Importance of effective writing, Paragraph writing, Email etiquettes.	4		4

Reference books:

1. Technical Communication by Meenakshi Raman and Sangeeta Sharma, OUP
2. Public Speaking and Influencing Men in Business by Dale Carnegie
3. Professional Communication Skills
by Bhatia and Sheikh, S. Chand Publications
3. Communication Skills by Sanjeev Kumar and Pushpalata, OUP
4. Communication Skills by Lalita Bisen, Bhumika Agrawal, N. Thejo Kalyani, Himalaya Publishing House

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Sem: I	Total Hours Distribution per week		
Total Credit :1	Practical (P): 1 Hrs.		
Subject Code	BAE1P01	Communication Skills	
Examination Scheme - Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	--

LIST OF PRACTICAL:

Pr. No.	List of Practical: (Perform any 6 – 8 Practical)
1	Barriers to Communication
2	Non-verbal Communication
3	Listening Skills
4	Reading Skills
5	Speaking Skills
6	Presentation Skills
7	Group Discussion
8	Interview Techniques
Pr. No.	List of Practical: Beyond Syllabus
1	Development of Word Power
2	Use of Figurative language

Suggested Textbooks/Reference Books/ Web page (URL)/Research paper, etc.

- 1 Technical Communication by Meenakshi Raman and Sangeeta Sharma, OUP
- 2 Public Speaking and Influencing Men in Business by Dale Carnegie
- 3 Professional Communication Skills by Bhatia and Sheikh, S. Chand Publications
- 4 Communication Skills by Lalita Bisen, Bhumiika Agrawal, N.Thejo Kalyani, Himalaya

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Sem: I	Total Hours Distribution per week		
Total Credit :2	Practical (P): 4 Hrs.		
Subject Code	BVS1P03	Basics of Solar PV Plant Installation	
Examination Scheme - Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
50 Marks	50 Marks	50Marks	--

Practical Course Objectives	
1	To understand the Solar PV Technology
2	To learn the basics of Solar PV Panels
3	To prepare the design of Solar PV Plant
4	To install a small Solar PV Plant

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Compute the electricity generation from solar radiations
2	Carry out basic requirement of solar PV plants
3	Identify and use the tools used for Solar PV system installation
4	Install the Civil/Mechanical and Electrical components of a Solar PV system
5	Test and Commission Solar PV system

LIST OF PRACTICAL:

Pr. No.	List of Practical
1	To demonstrate the I-V and P-V Characteristics of PV module with varying radiation and temperature level
2	To demonstrate the I-V and P-V characteristics of series and parallel combination of PV modules
3	To determine the different electrical parameters of a mono-crystalline and poly-crystalline silicon solar panel
4	To study the effect of variation in tilt angle on PV module power
5	To draw the charging and discharging characteristics of battery
6	To study solar PV inverter
7	Observe the output waveform of the inverter in auto mode
8	Workout power flow calculations of standalone PV system of load with battery

9	To study all the tools required for solar PV plant installation
10	To study the on-grid and off-grid Solar PV plant
11	To design a small Solar PV plant

Text & Reference Books:

1. Solar Photovoltaics: Fundamentals, Technologies and Applications by Chetan Singh Solanki, PHI
2. Install Your Own Solar Panels by Joseph Burdick, Philip Schmidt Storey Publishing
3. Solar PV Installer (Suryamitra) Published by RachanaSagar Pvt Ltd.

Online references:

1. The complete Solar Energy Course Beginner to Advanced level-
<https://www.udemy.com/course/the-complete-solar-energy-course-beginner-to-advanced-level/>

Second Semester Syllabus

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Sem: II		Total Hours Distribution per week	
Total Credit : 3		Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.
Subject Code		BEL2TO5	Engineering Mathematics -II
Examination Scheme			
Internal Marks:		University Marks:	Minimum Passing Marks:
30 Marks		70 Marks	45 Marks
			Examination Duration:
			3 Hours
Course Objectives			
1	The objective of the course is to inculcate and strengthen analytic ability among the engineering students and to create zeal of working with higher mathematics and its applications in the extensive field of engineering.		
2	The topics covered will serve as basic tools for specialized studies in many fields of engineering and technology.		

Course Outcomes	
After completion of syllabus, students would be able to	
1	Analyze real world scenarios to recognize when integrals are appropriate, formulate problems about the scenarios, creatively model these scenarios (using technology, if appropriate) in order to solve the problems using multiple approaches, judge if the results are reasonable, and then interpret and clearly communicate the results.
2	Define and understand the geometry of vector differential operators and line and surface integrals.
3	Explain and apply principles of study design and data collection.
4	Develop an ability to identify, formulate and/or solve real world problems
5	Understand the impact of scientific and engineering solutions in a global and societal context.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1 :Integral Calculus			
Evaluation of Definite and Improper Integrals: Beta and Gamma functions and their properties, Mean value, Mean square value and Root mean square	10		1

value, Tracing of curves (Cartesian), Applications of definite integrals to find length of curve, area, volume.			
Unit 2: Multivariable Calculus (Integration)			
Multiple Integration: Double integrals (Cartesian), Change of order of integration in double integrals, Change of variables (Cartesian to Polar). Applications on Area, Mass, Volume	10		2
Unit 3: Vector Calculus			
Vector Calculus: Vector triple product, Product of four vectors, Scalar point function, Vector point function, Vector differentiation, Gradient, Divergence and Curl, Directional derivatives, Solenoidal and Irrotational motions	8		3
Unit 4: Statistics			
Fitting of a Curve by Method of Least Squares: Straight line $y = a + b x$, Second degree parabola $y = a+bx+cx^2$ and curves of the type $y = ae^{bx}$, $y = ab^x$ and $y = ax^b$, Coefficient of correlation and Lines of regression, Rank correlation.	10		4
Unit-5: Numerical Methods:			
Error Analysis, Solution of Algebraic and Transcendental Equations: Method of False position, Newton–Raphson method and its convergence, Solution of system of simultaneous linear equations: Crout’s method (LU decomposition Method), Gauss-Seidel method.	8		5

Text/Reference Books:

- (1) Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
- (2) Ramana B.V., Higher Engineering Mathematics, Tata Mc-Graw Hill, New Delhi, 11th Reprint, 2010.
- (3) N.P. Bali and Manish Goyal, A textbook of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
- (4) B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
- (5) P. N. Wartikar and J. N. Wartikar, Applied Mathematics, Volume I and II.
- (6) H.K Dass, Rama Verma, Rajnish Verma, V.J. Dagwal, Sajid Anwar and D.F. Shastrakar, Engineering Mathematics, Volume I and II, S. Chand.

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Sem: II	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL2TO6	Applied Chemistry	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Students will be able to apply the basics concepts of electrochemistry& corrosion technology.
2	Students will know about fuels and lubricants and analyze the situation for their appropriate applications.
3	Students can analyze the various industrial problems arising due to water quality and their remediation.
4	Students will Develop the environmental awareness from the basics of green chemistry and its application.
5	Students will inculcate the use of instrumentation techniques and interpret its applications in material characterization.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1 :Electrochemistry and Corrosion Technology			
A] Electrochemistry: Electrochemical & Galvanic Series, Electrochemical & Electrolytic cell, Battery: Introduction, types, characteristics, components/materials, working and applications of Lithium-cobalt oxide and metal air batteries. Super capacitors: Introduction, types (EDLC, pseudo and asymmetric capacitor) with examples and applications. Energy conversion devices: Introduction, characteristics, materials, working and applications of H ₂ -O ₂ fuel cells, amorphous Si and quantum dye sensitized solar cells.	7		1

B Corrosion: Theories of Corrosion (Dry, Wet and Differential Aeration), Pilling-Bedworth Rule & Numerical, Factors affecting corrosion, Types of Corrosion (Intergranular & Stress), Corrosion Protection- Design & Material Selection, Cathodic Protection (Galvanic & Impressed Current)			
Unit 2: Fuels & Lubricants			
A Fuels: Introduction: Calorific value, Higher and lower calorific value; determination of calorific value by Bomb and Boy's calorimeter; numerical based on calorific value determination; Liquid fuels –fractional distillation of crude petroleum(boiling point wise separation only) use of gasoline and diesel in internal combustion engine: knocking and chemical constitution of fuel, Octane and Cetane number, doping agents, Introduction to propellants and its classification. Combustion calculations – Numericals based on combustion calculations for solid, liquid and gaseous fuels B Lubricants: Lubrication, Mechanism of lubrication, types of lubricants and its properties (viscosity & viscosity index, flash & fire point, aniline point, saponification value, acid value), criterion for selection of lubricants.	7		2
Unit 3: Water Technology			
A Water Purification Technology: Principles of coagulation and flocculation, Sterilization by using ozone and chlorine (Cl ₂ gas & chloramine), Break point chlorination and its significance. Industrial Water Treatment: Softening of water-principle- reactions, advantages, limitations and comparison of Zeolite process, and De-mineralization process. Numerical based on Zeolite process. B Boiler Troubles– Causes, effect on boiler operation and methods of prevention – Scales and sludges, Caustic embrittlement. Desalination of sea water- Principle, method and advantages of electro dialysis and reverse osmosis processes Waste Water Treatment (introduction and importance) –Water treatment from biological waste water to clean water production (Dissolved Air Floatation and Membrane bio reactors)	7		3
Unit 4: Green Chemistry			
A Green Chemistry: Introduction, twelve principles of Green chemistry with examples, Numerical based on atom economy, Carbon sequestration & Carbon Credits B Green reagents, Dimethyl carbonate and its applications, Supercritical carbon dioxide properties and applications Biopolymers: Classification based on type, properties and applications of collagen, chitosan, starch. Green Hydrogen Synthesis by photolysis of water, environmental benefits and applications	7		4
Unit-5: Material Characterization Techniques			

Principles and applications of – A] Electronic Spectroscopy (Beer-Lambert's law and its numerical), Infra-Red spectroscopy and Nuclear Magnetic Resonance spectroscopy. B] Thermal analysis (Thermogravimetry, Differential Thermal Analysis, Differential Scanning Calorimetry), Scanning Electron Microscopy, Transmission Electron Microscopy, Atomic Force Microscopy, Brunauer-Emmett-Teller (BET) surface area analysis, X-ray Diffraction Analysis, particle size analyser (Dynamic Light Scattering), High Performance Liquid Chromatography and Gas Chromatography	8		5
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References/ Text Books

1. Engineering Chemistry, S S Dara, S Chand Publication
2. Engineering Chemistry, Jain & Jain, Dhanpat Rai Publication
3. Applied Chemistry, A V Bharati, Das Ganu Publication
4. Energy & Environment, A V Bharati, Das Ganu Publication
5. Spectroscopy, Y R Sharma, S Chand
6. Green Chemistry for Beginners, Anju Srivastava, Rakesh K. Sharma, Jenny Stanford Publishing
7. Instrumental Methods of Chemical Analysis, B. K. Sharma, Krishna Prakashan.
8. Fundamentals of Solid Propellant Combustion, Kuo, K.K., Summerfield, M., Progress in Astronautics & Aeronautics, Vol. 90, AIAA. 1984 - https://onlinecourses.nptel.ac.in/noc24_ae09/preview

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Sem: II	Total Hours Distribution per week		
Total Credit :1	Practical (P): 2 Hrs.		
Subject Code	BEL2P06	Applied Chemistry Lab.	
Examination Scheme - Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
50 Marks	--	25 Marks	--

LIST OF PRACTICAL:

Pr. No.	List of Practical's (Any 6 performance based and 1 virtual lab experiment)
1	Proximate Analysis of coal
2	Estimation of viscosity of oil by Redwood Viscometer 1 or 2
3	Estimation of Flash point of lubricating oil by open/ closed cup apparatus
4	Estimation of Acid value of lubricant
5	Estimation of Consistency of grease by penetrometer
6	Estimation of Saponification value of lubricant
7	Estimation of Hardness of water (Total, Permanent & Temporary) by complexometry
8	Estimation of Alkalinity of water (Warder's Method)
9	Estimation of DO / free chlorine estimation
10	Estimation of Copper estimation (odometrically)
11	Estimation of Ni by complexometry / gravimetry.
12	Fe(II)/ (III) estimation by redox titration.
13	Beer's Law verification by spectrophotometer.
14	Separation of copper nickel ions by paper chromatography.
15	Acid base titration by potentiometry
16	Acid base titration by conductometry
17	Virtual Lab: Experiment on Calorimetry
18	Virtual Lab: Experiment on Spectroscopy

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Sem: II	Total Hours Distribution per week		
Total Credit:3	Lecture(L):3Hrs	Tutorial/Activity(T/A): Hrs.	Practical(P): Hrs.
Subject Code:	BEL2T07	Name of Subject: Engineering Mechanics	
Examination Scheme			
Internal Marks:	University Marks:	Maximum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objective	
1	To Understand the force systems and draw free body diagram to analyze rigid body equilibrium.
2	To Comprehend the principles of friction and solve engineering mechanics problems associated with frictional force.
3	To introduce the concept of centriod and moment of inertia of plane area.
4	To make student aware about the method of joint and method of section to analyse the truss.
5	To Understand the concept of motion of particles and rigid bodies.

List of Course Outcome	
After completion of syllabus student able to	
1	Analyse a given force system and apply the knowledge of force system and moment to determine resultant of various force system & apply the knowledge of equilibrium of force system.
2	To apply the basics of friction
3	Locate Centroid and compute area Moment of Inertia for standard shapes and composites areas.
4	Analyze simple determinate trusses
5	Analyzetheconnectedsystemofparticlesusingconceptofdynamicequilibrium

SYLLBUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
UNIT 1: Basics Concepts and equilibrium of force systems			1
Introduction to Engineering Mechanics: Force Systems, Basic concepts, Rigid Body equilibrium;	3		
System of Forces, Coplanar Concurrent Forces, Resultant Moment of 4 Forces and its Applications; Couples in 2D and 3D	4		

Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems. Simple beams and support reactions. Diagram of Statically Determinate Beams	3		
UNIT 2: Friction			
Types of friction, Limiting friction, Laws of Friction, Static Friction; Motion of Bodies, wedge friction, ladder friction Belt friction	5		2
UNIT 3: Centroid and Moment of Inertia			
Centroid and Centre of Gravity, Centroid of standard shapes and application to composite plane shapes; Area moment of inertia: formula of area moment of inertia of standard shapes, parallel and perpendicular axis theorem, application to composite plane shapes	7		3
UNIT 4: Analysis of structures- Trusses			
Assumption and Analysis of simple trusses by method of joints & method of sections	5		4
UNIT 5: Kinetics of Particles			
Kinetics of particles D'Alemberts principle and its application in connected system of particles, Impulse Momentum, Collision of bodies, Work Energy Method.	5		5

References/ Text Books

1. Vector Mechanics for Engineers: Statics and Dynamics, Johnston. R.E., Beer. F., Eisenberg. E. R,& Mazurek. D., McGraw Hill
2. Engineering Mechanics: Principles of Statics and Dynamics, R. C. Hibbler, Pearson Press
3. Engineering Mechanics, S S Bhavikatti, New-age International publication

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: II	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL2T08	Elements of Electrical Engineering & Measurements	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives	
1	To learn fundamentals of electrical circuits and analysis
2	To understand the basics of electrical machines
3	To learn the various electrical measuring instruments
4	To learn the concept of transducers

Course Outcomes	
After completion of syllabus, students would be able to	
1	Apply the concepts for analysis of single phase and three phase a.c. circuits
2	Understand the principle and working of electrical machines and their basic analysis
3	Learn the various a.c. bridges and their applications to measure R-L-C parameters
4	Understand the principle and working of electrical measuring instruments
5	Learn the concepts of transducers and their applications

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1			
A.C Circuits: Periodic Function, Average & R.M.S., Values, Steady State Behavior With Sinusoidal Excitation, Phasor Representation, Reactance & Impedance, Series & Parallel Circuit, Phasor diagram Power Factor, Principle of Generation of Single Phase & Three Phase Voltages, Power in Balanced Three Phase AC System	5		1
Unit 2:			
Principle and working of single phase transformer, transformers losses, phasor diagram, d.c. machines construction and working, simple numerical analysis	5		2
Unit 3:			
Measurement of low, high and medium resistances, Principle and working of A. C. Bridge, Various types of bridges and applications for measurement of R-L-C parameters	5		3
Unit 4:			

Classification of measuring instruments, Construction and working principle of moving coil, moving iron and dynamo meter type of instruments, measurement of power, energy, power factor	5		4
Unit-5:			
Fundamentals of Sensors and transducers, classification of transducers, transducer applications for measurement of displacement, pressure, temperature etc	4		5

Text & Reference Books:

1. S.K. Bhattacharya , “Basic Electrical and Electronics Engineering”, Pearson Education, 2012
2. D. C. Kulshrehtha, "Basic Electrical Engineering", Tata McGraw Hill, 2012
3. Kothari D.P. and Nagrath I.J., “Theory And Problems of Basic Electrical Engineering,” Prentice Hall
4. A.K. Sawhney, “A course in Electrical & Electronics Measurement and Instrumentation”, Dhanpat rai & Sons, 2015
5. Ernest O.Doebelin, “Measurement Systems Application and Design, International Student Edition”, McGraw Hill Book Company, 1998

Online references:

1. Basic Electrical Engineering A Web course of NPTEL by Day, Bhattacharya & Roy, Available:- [www. nptel.ac.in](http://www.nptel.ac.in)

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Sem: II	Total Hours Distribution per week		
Total Credit :1	Practical (P): 2 Hrs.		
Subject Code	BEL2P08	Elements of Electrical Engineering & Measurements	
Examination Scheme - Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25Marks	--

Practical Course Objectives	
1	To analyze electrical a.c. circuits
2	To understand the operating principles of electrical machines
3	To learn the use of electrical measuring instruments for measuring various parameters
4	To understand the working of transducers and their applications

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	To compute the various parameters of a.c. circuits
2	To perform the short circuit test and open circuit test for the transformer
3	To measure and compute the parameters of electrical machines
4	To understand the use of a.c. bridges for R-L-C measurements

LIST OF PRACTICAL:

Pr. No.	List of Practical
1	To study RLC series ac circuit
2	To study resonance in RLC series ac circuit
3	To study RLC parallel ac circuit
4	To study 3-phase circuits with load connected in star and delta
5	To determine Voltage regulation and efficiency of a single phase transformer by direct loading
6	Study of d.c. motors
7	Measurement of three phase power by Two Wattmeter method procedure
8	To study and plot the characteristics of LVDT
9	To analyse the characteristics of the Piezo electric sensor
10	To measure unknown values of given inductance and capacitance using appropriate a.c. bridge

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Sem: II	Total Hours Distribution per week		
Total Credit :2	Practical (P): 4 Hrs.		
Subject Code	BSE2P01	Electric Wiring & Illumination System	
Examination Scheme - Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
50Marks	50 Marks	50 Marks	--

Practical Course Objectives	
1	To develop a foundational understanding of electric charge
2	To equip students with the skills to proficiently install various wiring types
3	To cultivate competence in diverse lighting systems
4	To enable students to apply their knowledge in integrating lighting with smart home platforms designing energy-efficient solutions, and implementing sustainable lighting practices and maintenance plans.

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Solve basic problems in electrical systems using fundamental electrical concepts
2	Interpret electrical wiring diagrams and troubleshoot electrical issues
3	Understand various lighting sources and smart lighting systems
4	Apply knowledge to integrate lighting with smart home platforms and design energy-efficient solutions.

LIST OF TOPICS AND EXPERIMENTS:

Pr. No.	List of Topics and Experiments
1	Study of electrical systems fundamentals- Introduction to Electricity, Basics of electric charge, voltage, resistance, current and Ohm's Law, Overview of circuit components (resistors, capacitors, inductors), Series and parallel circuits, Energy consumption in electrical systems, Common safety practices in electrical systems, Introduction to personal protective equipment (PPE)
2	Study of electrical wiring systems- Overview of wiring types (conduit, non-metallic sheathed cable, armored cable), Reading and interpreting electrical wiring diagrams, Common symbols and conventions in electrical wiring diagrams, Guidelines for installation of electrical wiring, Basic troubleshooting techniques in electrical systems
3	Study of various Illumination Systems- Overview of lighting sources (incandescent, fluorescent, LED), Factors influencing lighting design (purpose, aesthetics, energy efficiency), Introduction to manual and automated lighting controls. Smart lighting systems and their benefits, LED technology and its advantages.

4	Design of smart home integration and sustainable lighting practices- Overview of home automation in the context of electrical systems and lighting, Integration with smart home platforms, Design of energy-efficient lighting systems, Routine maintenance practices for lighting systems, Development of lighting maintenance plan
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Text & Reference Books:

1. Handbook of Electrical Installation Practice by Geoffrey Stokes
2. Electrical Wiring Residential by Ray C. Mullin
3. Illumination and Electrification of Buildings by M. Chaudhari

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Sem: II	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BIK2T01A	Consciousness Studies	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Analyze the basics of Psychology and its applications
2	Develop knowledge about the sensory processes and perception
3	Apply various theories of classical conditioning
4	Integrate the theories of memory and behavior of mind

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1			
An introduction to Psychology Introduction to Psychology, Definition of psychology, history, methods in Psychology, Subfields of Psychology and its applications	10		1
Unit 2:			
Basic Cognitive Processes Sensory processes-general characteristics of senses, visual sense, auditory sense, other senses Perceptual organization-principles of perceptual organization, object perception and perceptual constancies, influences upon perception, extrasensory perception	10		2
Unit 3: Learning			
Classical conditioning, theories about classical conditioning, Reinforcement and Punishment	9		3
Unit 4: Memory			
Theories about memory, brain and memory, long term memory, forgetting	7		4

Reference Books:

1. Clifford T. Morgan, King, Weisz and Schopler, Introduction to Psychology, McGraw Hill Education (India) Private Limited
2. Hilgard, Atkinson and Atkinson(1977). Introduction to Psychology. Tata McGraw Hill
3. Kao H.S R.& Sinha D. (Eds)(1977). Asian perspectives on psychology. New Delhi: Sage

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Sem: II	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BIK2T01B	Preserving Art, Culture and Tradition	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives	
1	To provide overview of Indian Knowledge System (IKS) and sensitize the students to the contributions made by Indians in the field of philosophy, art and health.

Course Outcomes	
After completion of syllabus, students would be able to	
1	Interpret basics of Indian Knowledge system.
2	Integrate the teaching of Indian culture and civilization
3	Appreciate Indian artistic tradition.
4	Analyze Indian health and wellness system for healthy living

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1			
Introduction to Indian Knowledge System Introduction and overview of Indian Knowledge system, The Vedic Corpus -Vedas, Types of Vedas, Upavedas, Types of Upavedas	9		1
Unit 2:			
Indian Culture and Civilization Indian culture and Civilization: its characteristics, Difference between Culture and Civilization, Indus valley civilization, Vedic civilization	9		2
Unit 3:			
Indian Artistic Tradition, Indian Artistic tradition: Chitrakala- Indian style painting (Madhubani, Warli, Phad, Kalamkari, Gond, Mandana), Nritya : Indian dance forms (Bharatnatyam, Kathak, Kathakali, Kuchipudi, Manipuri, Mohiniattam) Sangeet- Carnatic music & Hindustani music	9		3
Unit 4:			
Health and Wellness Health and Wellness, Well being: Mental & Physical, Dimensions of Wellness, Concept of healthy living in Ayurveda, Tri-doshas –Relationship to Health	9		4

Activity: Prepare PPTs/Posters/Videos on any two topics

Books Recommended:

1. Introduction to Indian Knowledge System by Mahadevan, B, Bhat, VinayakRajat,NagendraPavana R.N., Prentice Hall India Pvt., Limited, 2022.
2. Indian knowledge Systems, Kapil Kapoor, Avadhesh Kumar Singh, D.K, Printworld.
3. Traditional Knowledge System in India by Amit Jha, Atlantic Publishers, 2002
4. Exploring The Mysterious, By T.N. Dhar · Mittal Publications, 2004
5. Indian Art & Culture (E), By Anurag Kumar, Arihant Publication India Limited, 2016
6. A History of Indian Philosophy, Volume 2, By SurendranathDasgupta, DiamondPublishers, 2017
7. Sri Suresh Soni, Sources of our cultural heritage, PrabhatPrakashan, 2018.
8. A Beautiful Tree by Dharampal, RashtrotthanaSahitya, 2021

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Sem: II	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BIK2T01C	Wellness, traditional medicines and yoga	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives	
1	The course will enable engineering students to acquire the knowledge of richness of healthy lifestyle and strong heritage of yoga and Vedas in Indian traditional system.

Course Outcomes	
After completion of syllabus, students would be able to	
1	Understand the importance of a healthy lifestyle
2	Familiarize to manage stress and health consciousness about physical and mental health.
3	Appreciate the benefits of yoga and medicinal plant.
4	Identify the social changes in Indian society.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1			
Importance of health and wellness, Essential components of balanced diet for healthy living, Processed foods and unhealthy eating habits.	9		1
Unit 2:			
Body systems and common diseases, Sedentary lifestyle and its risk of disease, Stress, anxiety, and depression, Factors affecting mental health.	9		2
Unit 3:			
Importance and benefits of yoga, Purpose of yoga, traditional knowledge of medicinal plant, use of home available herbs and spices.	9		3
Unit 4:			
Vedas and it types, Social change in Indian society, Social stratification and class conflicts.	9		4

Textbooks/References:

1. Sociology in India – Surendra Sharma, Rawat Publication.
2. Bradfird B, Strand and Others. Fitness Education Arizona Gorsuch Seani; sbrickPublishers, 1997.
3. Scott K. Powers and Stephen L. Dodd. Total Fitness: Exercise, Nutrition and wellness, Boston: Allyn and

Bacon, 1999.

4. Rigveda Samhita with Sayanabhasya, Vaidik Samshodhan Mandal, Pune

5. Riksuktashati, H. D. Velankar, Bharatiya Vidya Bhavan, Mumbai

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Sem: II	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BIK2T01D	Glimpses of ancient Science and Technology	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives	
1	To provide the students with scientific foundation of Ancient Indian Knowledge System
2	To create awareness about scientific heritage of the ancient civilization
Course Outcomes	
After completion of syllabus, students would be able to	
1	To provide information about great mathematicians and to help students to trace, identify, practice, and develop the significant Indian mathematics
2	To understand the concept of motion and its application in Indian ancient physics literature.
3	To understand the concepts of basic chemical & metallurgical process of ancient and medieval India.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-1			
Mathematics in India: Introduction of inception of Mathematics from vedic periods.Great Mathematician and their contribution (e.g. Arytabhatta, Bhaskara, Brahmagupta, Ramanujan,Pingala, Bhaskara-II), Sulbhasutras (Pythagoras theorem), Square, Square root, Square root of imperfect Squares, Magic Squares, Value of Pi.	12		1
Unit 2:			
Physics in India: Vaisheshikadarshan Atomic theory & law of motion, theory of Panchmahabhoota, BrihathShathaka (divisions of the time, unit of distance), Bhaskarachaya (Introduction to theory of Gravity, Suryasiddhanta&Sidhantashriomani), Lilavati (Gurutvakashan Shakti).	12		2
Unit 3:			
Chemistry in India: Vatsyayana, Nagarjuna, Vagbhata –building of Theras-Shala(laboratory), working arrangements of Ras-Shala, material and equipment,YaśodharaBhaṭṭa-processof distillation, apparatus.Metallurgy in India: Survarṇa(gold) and its different types, properties, Rajata(silver),Tamra(copper), Loha(iron), Jasta(zinc), Naga /Sisa(lead), Pittala(brass).	12		3

Reference Books Recommended:

1. Kapur K and Singh A.K (Eds) 2005). Indian Knowledge Systems, Vol. 1. Indian Institute of Advanced Study, Shimla. Tatvabodh of Sankaracharya, Central Chinmay Mission Trust, Bombay,1995
2. Dharmpal, Indian Science and Technology in the eighteen century, Rashtrottahanasahitya, 1983
3. S Biswal, B L Ray, Vedic Science and technology, DK Print world, 2009
4. A.K Bag, History of technology in Indian (Set 3 vol), Indian Nation Science Academy, 1997.
5. A Gosh, History of Science in India (Volume-I Part-II Astronomy), the national academy ofscience, India & the Ramkrishna mission institute of culture, 2014

Third Semester Syllabus

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Sem: III	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL3T09	Network Analysis	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Understand and analyze basic circuit elements and their interconnections
2	Apply mesh current and node voltage methods to analyze electrical circuits.
3	Obtain transient and steady-state responses of electrical circuits.
4	Synthesize waveforms and apply Laplace transforms to analyze networks
5	Evaluate different Network Functions and understand two port network behavior

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I : Electric Circuit Elements			
Basic concepts: Ideal source, Practical source, dependent and independent source Voltage and Current sources, source transformation, concept of duality, Series and Parallel Combination of Inductors , Series and Parallel Combination of Capacitors.	7		1
Unit –II: Nodal and mesh Analysis			
Introduction, mesh analysis ,the super mesh, mesh basis equilibrium approach for simple networks of having mutual coupling. Nodal analysis, the super node, nodal vs mesh analysis	7		2
Unit –III: Network Theorems			
Superposition, Thevenin's, Norton's, Maximum Power transfer, Reciprocity theorems as applied to D C. & A.C	7		3
Unit –IV: Electric Circuit Analysis using Laplace Transforms			
Review of Laplace transform, waveform synthesis, Analysis of electrical circuits using Laplace transform for standard inputs, Solution of first and second order differential equations of different combinations of series and parallel RLC networks. analysis of networks with and without initial conditions using Laplace transforms.	7		4

Unit –V: Two port networks and Network functions			
Two port networks, relationship between two port variables, driving point and transfer functions, properties, concept of complex frequency, Poles and zeros. Two port network parameters: Impedance parameters, admittance parameters, transmission parameters and hybrid parameters, interconnection of two port networks	8		5

Text Books:

1. Van Valkenburg, “Network Analysis”, Third Edition, 2009, Prentice Hall of India
2. Sudhakar, A, Shyammoan, “Circuits and Networks”, Third Edition, 2006, Tata McGraw-Hill.
3. D. Roy Choudhary, “Networks and Systems”, New Age International Publishers, 2nd Edition, 2012
4. Kelkar and Pandit, “Linear Network Theory”, Pratibha Publications.

Reference Books:

1. Mahmood Nahvi, Joseph A Edminister, “Schaum’s outline of Electric Circuits”, 6th Edition, Tata McGraw-Hill, 6th Edition, 2013
2. W. H. Hayt and J. E. Kemmerly, “Engineering Circuit Analysis”, McGraw Hill Education, 2013.
3. C. K. Alexander and M. N. O. Sadiku, “Electric Circuits”, McGraw Hill Education, 2004.
4. K. V. V. Murthy and M. S. Kamath, “Basic Circuit Analysis”, Jaico Publishers, 1999.
5. K. Sureshkumar, “Electric Circuits & Network”, Pearson Publication
6. Del Toro, “Electrical circuit”, Prentice Hall

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Sem : III	Total Hours Distribution per week		
Total Credit:1	Practical(P): 2Hrs.		
Subject Code	BEL3P09	Network Analysis Lab	
Examination Scheme- Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
Marks	Marks 25	Marks 25	--

Practical Course Objectives	
1	Understand and Apply Network Theorems
2	Analyze Resonance in Electrical Circuits
3	Evaluate Network Parameters
4	Enhance Circuit Analysis and Problem-Solving Skills
5	Bridge Theoretical Knowledge with Practical Applications

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Ability to Apply Network Theorems
2	Understanding of Resonance in RLC Circuits
3	Proficiency in Two-Port Network Parameter Analysis
4	Practical Skills in Electrical Circuit Design and Testing
5	Enhanced Problem-Solving and Analytical Skills

LIST OF PRACTICALS:

Pr. No.	List of Practical's
1	To study and verify the Superposition Theorem.
2	To study and verify the Thevenin's Theorem.
3	To study and verify the Reciprocity Theorem.
4	To study and verify the Maximum Power Transfer Theorem.
5	To study and verify the Nortons Theorem.
6	To study and verify the Tellegens Theorem.
7	To study the resonance in RLC Series Circuit and find its resonant frequency.
8	To study of the resonance in RLC Parallel Circuit and find its resonant frequency.
9	To determine Y-parameter for π network.
10	To determine Z-parameter for T network
11	To determine ABCD Parameter of a Passive Two Port Network.1)
12	To determine h-parameters for a two-port network.

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Sem: III	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BOE3T010	RENEWABLE ENERGY SOURCES	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes

After completion of syllabus, students would be able to

1	Memorize the fundamental of solar radiation geometry
2	Identify and analyze the process of power generation through solar photovoltaic
3	Highlighting the various applications of Solar Energy
4	Outline the site requirement criteria for wind farm & compare different types of wind generators.
5	Identifying non-conventional Energy sources such as Geothermal, MHD, Biomass, Fuel cell, Tidal, Ocean for generating Electricity

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit 1: Solar Radiation & its Measurement			
Solar Radiation & its Measurement: Solar Constant, Solar radiation at earth's surface, solar radiation geometry, solar radiation measurement, estimation of average solar radiation, solar radiation on tilted surfaces	6		1
Unit 2: Solar Photovoltaic power generation			
Solar Photovoltaic power generation: Physics of solar cells, Characteristic of solar cell, series and parallel connection, types of solar cell, module manufacturing, partial shading, bypass and blocking diode, load calculation, different panel calculations and selection (Monocrystalline, Polycrystalline etc), Calculation of Solar rooftop setup (rating): stand alone PV system with battery and grid connected PV system with Net Metering, Introduction to MPPT	6		2
Unit 3: Application of Solar Energy			
Application of Solar Energy: Solar water heating, space heating, space cooling, solar thermal heat conversion, Solar Cooking, Solar pumping, Solar Water	6		3

pumping for agriculture purposes, Calculation of solar setup required in solar water pumping, Solar Green Houses, Hydrogen production from Solar Energy			
Unit 4: Wind Energy			
Basic principles of wind energy conversion, wind energy conversion system, wind data & energy estimation, site selection consideration, basic components of wind energy conversion system (WECS), classification of WEC system, generating system, energy storage, application of wind energy. Stand-Alone and Grid Connected Wind-Electrical Power System	6		4
Unit 5: Other Nonconventional Energy Source			
Brief Introduction to operating principles only: MHD power generation, Fuel cell, Energy from Ocean, Ocean thermal electric conversion (OTEC), Claude & Anderson cycles, Hybrid cycle, Energy from Tides ,Estimation of Energy & Power in simple single basin ,Tidal system			

Suggested Text Books &Reference Books:

- 1 Non Conventional Energy Sources G.D. Rai, Khanna publishers
2. Non Conventional Energy Resources B. H. Khan 2nd , The McGraw Hill Companies
3. Solar Energy: Principles of thermal collection and storage, S. P. Sukhatme 2nd edition, Tata McGraw Hill Publishing Company Ltd.
4. Solar Photovoltaics: Fundamental, Technologies and Applications, Chetan Singh Solanki ,
5. Non-Conventional Energy Sources and Utilization, R.K. Rajput, S. Chand Publications.
6. Non-Conventional Energy Resources, D S Chauhan, S K Srivastava, New Age International Publishers

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Sem:	Total Hours Distribution per week		
Total Credit:1	Practical(P):2 Hrs.		
Subject Code	BOE3P10	RENEWABLE ENERGY SOURCES LAB	
Examination Scheme- Practical			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
Marks: 25	Marks 25	Marks 25	--

Practical Course Objectives	
1	To analyze the impact of shading on the output power of a photovoltaic (PV) system.
2	To demonstrate the working of bypass and blocking diodes in a photovoltaic system.
3	To calculate and analyze the power flow in a standalone PV system supplying DC and AC loads with a battery.
4	To study the performance of MPPT through varying resistive loads and DC-DC converter duty cycles.

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Students will be able to identify and describe the effects of shading on the output power of PV modules.
2	Students will demonstrate an understanding of the function of bypass and blocking diodes in PV systems.
3	Students will apply power flow calculations and analyze the operation of a standalone PV system providing DC and AC loads, integrating a battery.
4	Students will evaluate the performance of a PV system by studying MPPT techniques, comparing results obtained through varying resistive loads and DC-DC converter duty cycles.
5	Students will analyze the behavior of boost and buck converters and their role in power conversion within a PV system.

LIST OF PRACTICALS:

Pr. No.	List of Practical
1	To demonstrate the effect of shading on module output power.
2	To demonstrate the working of diode as Bypass diode and blocking diode.
3	Workout power flow calculations of standalone PV system of DC and AC load with battery.
4	Find the MPP manually by varying the resistive load across the PV panel.
5	Find the MPP by varying the duty cycle of DC-DC converter.
6	Observe the V_m , I_m , P_m and duty cycle at which MPP occurs, with MPP algorithm
7	Observe the response of P_m in the plotter and compare with the P_m observed.
8	Study of various modes of Constant Voltage Charging technique.
9	Study of Boost converter
10	Study of Buck converter
11	Study of Boost converter
12	Study of Dusk to Dawn switch

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Sem: III	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BMD3T11	ELECTRONICS DEVICES & CIRCUITS	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Understand the characteristics of basic diodes and analyze rectifier circuits
2	Understand the characteristics and use of a transistor.
3	Apply the knowledge of transistor for the analysis of power amplifiers and oscillators.
4	Evaluate the characteristics of basic electronic devices and analyze emerging technologies in renewable energy systems and energy storage solutions.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit 1: Basic diode and its applications			
P-N junction diode, operation and characteristics; half-wave and full-wave rectifiers, Filters, Ripple factor, characteristics and applications of Zener diodes, Zener diode voltage regulator, photo diodes.	7		1
Unit 2: Basic transistor and its applications			
Operation and characteristics of a BJT. BJT as a switch. BJT as an amplifier: Biasing circuits, small-signal analysis of CE, CB and CC amplifiers and their Comparison, Transistor as a switch. Field effect transistors and MOSFETs- Principle of operation .and characteristics.	10		2
Unit 3: Power amplifiers			
Classification as A, B, AB, C, Push pull amplifiers, Cross over distortion, Positive and Negative amplifiers- classification, feedback amplifiers, advantages and applications. Differential amplifier circuits and their stages, current source, biasing, level Shifting techniques, Oscillators- Bark hausen's criterion, RC and Crystal oscillators	10		3

Unit 4: Applications of Op-Amp			
Inverting and non-inverting amplifier, integrator, active filter, voltage regulator, oscillators (Wein's bridge and phase shift). Analog to Digital Conversion. Hysteresis Comparator, Zero Crossing Detector, Square-wave and triangular-wave generators. Precision rectifier. Study of linear ICs: LM741, LM555, LM 7805	08		4

Suggested Text Books:

1. Millman and Halkias, "Electronic Devices and Circuits", McGraw Hill.
2. Millman and Halkias, "Integrated Electronics", McGraw Hill
3. J. V. Wait, L. P. Huelsman and G. A. Korn, "Introduction to Operational Amplifier theory and applications", McGraw Hill U. S., 1992.
4. R. Gaikwad, "Operational Amplifiers and applications"

Suggested Reference Books:

1. J. Millman and A. Grabel, "Microelectronics", McGraw Hill Education, 1988.
2. P. Horowitz and W. Hill, "The Art of Electronics", Cambridge University Press, 1989.
3. P. R. Gray, R. G. Meyer and S. Lewis, "Analysis and Design of Analog Integrated Circuits", John Wiley & Sons, 2000

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
NAGPUR**

**FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: III	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BOE3T01	Fundamental of Electrical Engineering	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes

After completion of syllabus, students would be able to

1	Analyze DC circuits using fundamental electrical concepts, Ohm's law and Kirchhoff's laws, and solve problems involving basic electrical components such as resistors, capacitors, and inductors.
2	Apply the principles of AC circuit analysis to calculate power factors, understand phasor representations, and evaluate the behavior of RLC circuits, including resonance conditions.
3	Explain the working principles of electrical machines and transformers, and evaluate their applications in power systems while considering energy efficiency aspects.
4	Evaluate the characteristics of basic electronic devices and analyze emerging technologies in renewable energy systems and energy storage solutions.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit 1: Basic Electrical Concepts and DC Circuits			
Introduction to electrical engineering and its applications, Electric charge, current, voltage, power, and energy, Electrical components: resistors, capacitors, inductors Ohm's law and Kirchhoff's laws, Series and parallel circuits, DC circuit analysis techniques	6		1
Unit 2: AC Circuits and Power Systems			
Alternating current fundamentals , Sinusoidal waveforms and phasors, RLC circuits and resonance ,Power in AC circuits: real, reactive, and apparent power* Power factor and power factor correction.	6		2
Unit 3: Electrical Machines and Transformers			
Magnetic circuits and electromagnetic principles , Transformers: principle, types, and applications, DC machines: motors and generators, AC machines: induction motors, synchronous machines, Speed control techniques, Introduction to electric drives	6		3

Energy efficiency in electrical machines			
Unit 4: Electronics devices and Emerging Technologies			
Semiconductor devices: diodes, transistors, Power electronic devices and basic circuits Operational amplifiers and their applications ,Renewable energy systems: solar, wind Energy storage technologies	6		4

Suggested Text Books &Reference Books

1. Fundamentals of Electrical Engineering and Electronics" by S.K. Sahdev , Dhanpat Rai & Co.
2. Basic Electrical Engineering" by V.N. Mittle and Arvind Mittal , Tata McGraw Hill
3. Basic Electrical and Electronics Engineering" by R.K. Rajput , Laxmi Publications
4. Fundamentals of Electrical Engineering" by B.L. Theraja and A.K. Theraja S. Chand Publishing
5. Basic Electrical Engineering" by D.P. Kothari and I.J. Nagrath , McGraw Hill Education

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
NAGPUR**

**FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: III	Total Hours Distribution per week		
Total Credit : 1	Practical (P): 2 Hrs		
Subject Code	BOE3P01	Fundamental of Electrical Engineering LAB	
Examination Scheme			
Internal Marks	University Marks	Minimum Passing	Examination Duration
25 Marks	25 Marks	25 Marks	---

Practical Course Objectives	
1	Analysis and Implementation of DC Circuits
2	Analyze AC Circuits and Power Measurement
3	Testing and Performance Analysis of Electrical Machines & Transformers
4	Understand Application of Semiconductor Devices and Technologies

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Analyze DC Circuits using Ohm's and Kirchhoff's laws, measuring voltage, current, and resistance.
2	Study AC Circuits by observing waveforms, determining resonance, and measuring power parameters.
3	Test Electrical Machines through transformer tests, motor load tests, and performance analysis.
4	Implement Electronic Circuits including rectifiers, op-amp applications, and renewable energy systems.

LIST OF PRACTICAL:

Pr. No.	List of Practical
1	To Study and verification of Kirchhoff's Voltage and Current Laws in a DC Circuit.
2	To Study and verification of Superposition theorem.
3	To Study of RLC Circuits and Determination of Resonance Frequency
4	To Study RLC series circuit and to plot Phasor Diagram for it.
5	To Study & plot B-H curve for given magnetic material.
6	To Study and perform Direct loading test on single phase transformer.
7	To Study Half and Full wave rectifier.
8	To Study Characteristics of P-N Junction Diode
9	To Study Transistor Characteristics.
10	To Study Performance Evaluation of a Solar Panel System Under Different Load Conditions

Rashtrasant Tukadoji Maharaj Nagpur University B.Tech (EP/CE/CS/IT/CIVIL/ET) III Semester			
Course Title	Entrepreneurship and Startups	Course Code	BHM3T01
Teaching Hours/ Week	02 Th	Total Credits:	02
CIE	15 Marks	SEE	35 Marks

Course Outcomes:	
After completing the course, students will be able to	
CO1	Understand a know-how on entrepreneurship development
CO2	Acquire the knowledge of various types of startups
CO3	Remember the concept of ideation
CO4	Apply the funding for startups

UNIT 1
Concept of entrepreneurship, characteristics of an Entrepreneur, types of Entrepreneurship, Functions of Entrepreneurs, Women entrepreneurship in India, Problems and challenges of women entrepreneurs, Government's support system to develop women entrepreneurship.

UNIT 2
Concept of startup, Types of startups: Scalable startup, small business startup, lifestyle startup, buyable startup, social startup, big business startup, Startup ecosystem

UNIT 3
Concept of ideation, ideation process, idea incubation, design thinking approach, ideation techniques (brainstorming, sketching, SCAMPER, and prototyping), success factors for ideation.

UNIT IV
Funding for startups, angel funding, venture funding, difference between angel and venture funding, private equity fund, ownership of startups, causes of startups failures, Startup success case studies: Instagram, LinkedIn, Snapchat, Whatsapp

TEXT BOOK REFERENCE
1. Entrepreneurial Development By, S. S. Khanka S. Chand & Co. Ltd. New Delhi, 1999. 2. Entrepreneurial Development. By, S.Anil Kumar. New Age International. 3. Small- Scale Industries and Entrepreneurship, By, Dr. Vasant Desai, Himalaya Publication. 4. Industrial Economics and Entrepreneurship development by A.M. Sheikh, Nawaz Khan & M.A. Tongo, S chand Publication

Rashtrasant Tukadoji Maharaj Nagpur University B.Tech (Common to all branches) III Semester			
Course Title	Constitution of India	Course Code	BVE3T01
Teaching Hours / Week	02 Th	Total Credits:	02
CIE	15 Marks	SEE	35 Marks

Course Outcomes :	
After completing the course, students will be able to	
CO1	Analyze the basic structure of Indian Constitution.
CO2	Remember the Fundamental rights and duties.
CO3	Know DPSP's and Nation's political structure.
CO4	Understand function of Parliament and Judiciary.

UNIT 1
Introduction to the Indian Constitution, Preamble of the Indian Constitution and key concepts, Salient features of the Indian Constitution, Role and objective of Constituent Assembly.

UNIT 2
Fundamental rights meaning, significance, restrictions and limitations Fundamental duties and its scope, difference between Fundamental rights and Fundamental duties

UNIT 3
Directive Principles of State Policy (DPSP's) and its present relevance in India, Union Executive- President, Prime Minister and Union cabinet.

UNIT IV
Parliament - role and function, Lok Sabha and Rajya Sabha, Judiciary system in India, Supreme Court of India and other courts.

TEXT BOOK REFERENCE
1. Introduction to the Constitution of India by D D Basu. 2. Outlines of Indian Legal and Constitutional History by M P Jain. 3. Constitution of India by P M Bakshi

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: III	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): NA	Practical /Activity (T/A): 4 Hr.	
Subject Code	BCE3P01	Community Engineering Project	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
100 Marks	N.A	50 Marks	---

Project Objectives

1. Address real community needs through electrical engineering solutions
2. Develop practical implementation skills
3. Promote sustainable and cost-effective solutions
4. Foster community engagement and social responsibility
5. Apply theoretical knowledge to solve real-world problems

Project Guidelines

Phase 1: Community Need Assessment

- Conduct community surveys and interviews
- Identify specific electrical/energy-related problems
- Document existing infrastructure and limitations
- Assess available resources and constraints
- Consider socio-economic factors

Phase 2: Project Planning

- Define clear project objectives and scope
- Create detailed technical specifications
- Develop budget and resource requirements
- Establish timeline and milestones
- Form community partnerships
- Consider safety regulations and standards

Phase 3: Implementation

- Create detailed design documentation
- Source materials and equipment

- Execute project in phases
- Involve community members in implementation
- Document progress and challenges
- Conduct regular safety checks

Phase 4: Evaluation and Sustainability

- Measure project impact
- Gather community feedback
- Train community members for maintenance
- Document lessons learned
- Create maintenance schedules
- Prepare handover documentation

Sample Project Ideas

1. Solar-Powered Community Learning Center

Objective: Establish a reliable power supply for a rural community center

Components:

- Solar panel installation (1-2 kW system)
- Battery storage system
- LED lighting system
- Computer charging stations
- Basic electrical outlets

Expected Impact:

- Extended study hours for students
- Digital literacy programs
- Community gatherings after dark
- Emergency charging facility

2. Smart Street Lighting System

Objective: Implement energy-efficient street lighting in a residential area

Components:

- LED street lights with motion sensors
- Automatic timing controls
- Solar integration where possible
- Remote monitoring system
- Fault detection system

Expected Impact:

- Reduced energy consumption
- Improved street safety
- Lower maintenance costs
- Better lighting management

3. Micro-hydro Power Generation

Objective: Harness local water resources for power generation

Components:

- Small-scale turbine system
- Power distribution network
- Control and monitoring system
- Safety mechanisms
- Grid connection (optional)

Expected Impact:

- Clean energy generation
- Reduced power costs
- Local job creation
- Sustainable power source

4. Energy Audit and Efficiency Program

Objective: Improve energy efficiency in community buildings

Components:

- Energy monitoring systems
- LED lighting upgrades
- Power factor correction
- Smart metering
- Automated controls

Expected Impact:

- Reduced energy bills
- Better power quality
- Environmental benefits
- Community awareness

Documentation Requirements

Technical Documentation

1. Project proposal
2. Technical specifications
3. Circuit diagrams
4. Installation guides
5. Safety protocols
6. Maintenance manuals

Social Documentation

1. Community feedback forms
2. Impact assessment reports

3. Training materials
4. Sustainability plans
5. Photo and video documentation

Evaluation Criteria

Technical Aspects (40%)

- Design effectiveness
- Implementation quality
- Technical innovation
- Safety considerations
- Reliability

Social Impact (30%)

- Community benefit
- User adoption
- Sustainability
- Knowledge transfer
- Local capacity building

Project Management (30%)

- Planning and organization
- Resource utilization
- Timeline adherence
- Documentation quality
- Team collaboration

Tips for Success

1. Prioritize community involvement from the start
2. Use locally available materials when possible
3. Design for easy maintenance
4. Consider scalability and replication
5. Document everything thoroughly
6. Plan for long-term sustainability
7. Build strong community partnerships
8. Include training and capacity build.

Fourth Semester Syllabus

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: IV	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL4T11	Electrical Machine-I	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Determine Equivalent Circuit parameter, Efficiency and Regulation of Single Phase Transformer and to Explain the Phasor groups of Three Phase Transformer.
2	Analyze different characteristics of D. C. Motor and Speed Control of D.C. Motor.
3	Explain different types of Three Phase Induction Motor and Analyze the characteristics at different Value of Slip.
4	Know Voltage Regulation of Three Phase Synchronous Generator and Behaviour of Synchronous Motor with Different Excitations
5	Understand Single Phase Machines and Special Machines.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Single Phase Transformer			
Revision of Single-Phase Transformer, Phasor Diagram Under Different Load Conditions, Losses, Equivalent Circuit, Open Circuit and Short Circuit Test, Voltage Regulation, Efficiency, Condition of Maximum Efficiency, All Day Efficiency, Polarity Test. Single phase Autotransformer, Working, Merits and Demerits. Applications. Three Phase Transformer: -Principle and Operation, Connection and Phasor Groups, Polarity Test, Open Circuit and Short Circuit Test, Conditions of Parallel Operation.	8		1
Unit II: D.C. Machines			

Basic Principle and Operation of D.C. Motor and D.C. Generator, Emf Equation and Torque equation, Types of D.C. Machines, Characteristics and Speed Control of D.C. Shunt and D.C. Series Motor, Losses and Efficiency of D.C. Motor. Necessity of Starter and Constructional Details of Three Point Starter. Armature Reaction in D. C. Machines. Applications.	7		2
Unit III: Three Phase Induction Motor			
Construction Details, Types, Principle, Production of Torque, Torque Equation and Condition of Maximum and Starting Torque, Losses and Efficiency, Torque-Slip Characteristics, Behaviour for Different values of Slip. No Load Test and Blocked Rotor Test. Starting methods of Three Phase Induction Motor. Applications.	7		3
Unit IV: Synchronous Machines			
Three Phase Synchronous Generator : -Introduction, Constructional features of Salient Pole and Cylindrical Pole Rotor Machines, Introduction to Armature Winding and Field Winding, Winding Factors and EMF Equation, Armature Reaction, Phasor Diagram Under Load Condition, Regulation and Synchronous Impedance Method to Find Voltage Regulation. Three Phase Synchronous Motor: - Construction and Principle, Starting of Synchronous Motor, Motor on Load, Effect of Changing Field Excitation at Constant Load, V and Inverted-V Curves. Applications.	7		4
Unit V: Single Phase Machines			
Single Phase Induction Motor: - Principle and Operation, Double Field Revolving Theory. Principle and Working of Shaded Pole Induction Motor, Split Phase Induction Motor and Capacitor Start Capacitor Run Motor. Applications. Principle, Working and Applications of Special Machines: - Universal Motor, Hysteresis Motor, Brushless D. C. Motor, A.C. Series Motor.	7		5

Suggested Text books & Reference Books

Text Books

1. “Electrical Machines,” I. J. Nagrath, D.P. Kothari, Tata McGraw- Hill Publishing Company Ltd.
2. Electrical Machinery”, P.S. Bhimbra,, Khanna Publishers.
3. “Electrical Machines”, P.K. Mukherjee, S. Chakraborty, Dhanat Rai Publications.
4. “Generalized Theory in Electrical Machines” P.S. Bhimbra, Khanna Publishers.
5. “Basic Electrical Engineering,” D C Kulshreshtha, The McGraw Hill Higher Education Private Limited, New Delhi.
6. “Laboratory Courses in Electrical Engineering,” S.G.Tarnekar, P.P. Kharbanda, S.B.Bodkhe, S.D. Naik , S. Chand & Company Ltd., New Delhi
7. Use ICT Tools

Reference Books

1. “Performance and Design of A.C. Machines,” M.G.Say, CBS Publishers and Distributors Pvt. Ltd.
2. “Electrical Machinery”, A.F. Fitzgerald, Charles Kingdey, Jr. Stephan D. Umans, Fifth Edition in SI Units, McGraw Hill Book Company.
- 3., “Laboratory Manual for Electrical Machines, ”D.P. Kothari, B.S.Umre, Second Edition , I.K. International Publishing House Pvt.Ltd., New Delhi.

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: IV	Total Hours Distribution per week		
Total Credit : 1	Lecture (L): 0	Practical: 2 Hrs.	
Subject Code	BEL4P11	Electrical Machine-I LAB	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	-----

Course Objectives

1.	The basic principle of operation of 1- phase transformer, construction of 3-phase transformers, their classification, connections and phasor diagrams
2.	The basic principle, construction, operation, Performance characteristics, steady state analysis and applications of all types (AC and DC) electrical motors.

Course Outcomes

After completion of syllabus, students would be able to

1	Evaluation of performance indices, equivalent circuit parameters and variation of the performance on loading of single phase and three phase transformer, induction motors and DC shunt motors
2	Magnetizing characteristics of the DC Generator, Critical Field resistance and speed of the machine
3	Various methods of speed control of DC shunt motor and three phase wound rotor induction motor

Minimum Eight Practical's are to be performed out of the following:

Experiment No	Experiment Based on
1	To perform Open Circuit and Short Circuit tests on a 1-phase transformer to evaluate efficiency and voltage regulation.
2	To perform load test on 1-phase transformer to determine its efficiency and voltage regulation.
3	To study conversion of a 2-winding transformer into an autotransformer.

4	To study phasing out and polarity marking of a 1-phase transformer.
5	To study voltage and current relations in a 3-phase, Delta-Star connected transformer.
6	To perform Open Circuit and Short Circuit test on a 3-phase transformer.
7	To plot magnetization characteristic of a DC generator.
8	To study speed control of a DC shunt motor by varying – (a) field excitation and (b) armature voltage
9	To perform load test on a DC shunt motor.
10	To study control of a 3-phase slip-ring induction motor by – (a) variation of a rotor resistance and (b) varying supply voltage
11	To perform load test on a 3-phase induction motor by direct loading.
12	To perform load test on a 3-phase induction motor by indirect loading
13	To perform No-Load and Blocked rotor tests on a 3-phase induction motor
14	To Study starters used for starting AC and DC machines
15	Virtual Laboratory Experiments based on the above

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: III	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BEL4T12	POWER SYSTEM-I	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Understand the basic structure of power system, smart grid and micro grid.
2	Model and represent the power system components in it's per unit value.
3	Learn the parameters of transmission lines and cables.
4	Evaluate the performance of transmission lines.
5	Acquaint with the method of load flow analysis and the concept of voltage stability

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit 1: Evolution of Structure of Power Systems			
Structure of power systems, brief exposure to generation, transmission and distribution aspects, Present-Day Scenario, Introduction to Smart Grids and Micro-grids, their components, Standardization of transmission voltages, Overhead and Underground transmission system, EHVAC versus HVDC transmission, HVDC Components, distribution connection scheme(radial ,ring main and interconnected), Feeders and distributors, Substation and its equipment's.	8		1
Unit 2: Per Unit Representation			

Representation of power system elements, models and parameters of generator, transformer and transmission lines and load, voltage and frequency dependence of loads, single line impedance diagram, advantages of per unit representation.	6		2
Unit 3: Overhead Transmission Lines and Cables			
Components of overhead lines, choice of conductors, Skin effect, Proximity effect, Corona, Transposition of conductors, Bundled conductor, Types of insulators, string efficiency, Method to improve string efficiency, Derivation for Inductance of a single phase line, concept of self GMD and mutual GMD, Derivation for Capacitance of a single phase line, Insulated Cables, Dielectric stress in single core cables, Grading of cables, XLPE cables	8		3
Unit 4: Performance of Transmission line			
Classification of transmission line (short, medium (nominal T and nominal π) and long), Characteristics (voltage regulation and efficiency) of transmission lines, determination of generalized (A,B,C,D) constants for transmission line, Ferranti effect, Surge Impedance Loading, Series and Shunt Compensation of transmission lines (using capacitors only) .	8		4
Unit 5: Load Flow Studies			
Introduction to load flow studies, Classification of buses, Formation of bus admittance matrix, Static load flow equations, Gauss Seidel and Newton-Raphson method for solution (Numerical is not expected), Introduction of frequency and voltage as system state indicators, Concept of Voltage Stability, P-V and V-Q curves, Methods to improve voltage stability.	6		5

Text Book

1. I.J.Nagrath,D.P.Kothari,PowerSystemEngineering,TataMcGraw-Hillpublications, 2008
2. AshfaqHusain,ElectricalPowerSystem,CBSPublication,5thEdition
3. C.L.Wadhwa,ElectricalPowerSystems,NewAgeInternationalPubliser,6thEdition
4. V.K.MehtaandRohitMehta,PrinciplesofPowerSystem,S.ChandPublication, 2008

Reference Books:

1. W.D.Stevenson, Elements of power system analysis, McGraw-Hill publications,3rdEdition
2. O.I.Elgerd,ElectricEnergySystemsTheory:AnIntroduction,McGraw-Hill publications, 2ndEdition
3. Hadi Saadat, Power System Analysis,TMH, 2002
4. James A Momoh, Smart Grid:FundamentalsofDesignandAnalysis,Wiley2012
5. Janaka Ekanayake, Nick Jenkins, Kithsiri Liyanage,“Smart Grid: Technology and Applications”, Wiley 2012

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: IV	Total Hours Distribution per week		
Total Credit : 1	Practical (P): 2 Hrs		
Subject Code	BEL4P12	POWER SYSTEM LAB	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	--

Course Outcomes	
After completion of syllabus, students would be able to	
1	Determine various parameters of transmission line
2	Analyze the performance insulators (Liquid insulating medium & String efficiency of disc type solid insulator)
3	Understand the effects on transmission lines
4	Understand different types of faults
5	Analyze the characteristics of different types of relay

List of Experiments	
1	To determine A, B, C, D parameters of short and medium transmission line.
2	To study and compare the performance of ring and radial distribution system
3	Determination of power angle characteristics of transmission line
4	To study series impedance loading of transmission line
5	To draw PV characteristics of transmission line
6	To study the performance of oil testing set

7	To find out the string efficiency across the string of insulators
8	To study Ferranti effect and Corona effect on transmission line
9	To perform symmetrical fault analysis in AC network analyser.
10	To perform symmetrical fault analysis in DC network analyser & perform the experiment for Unsymmetrical fault analysis on DC network.
11	To study the characteristics of the operation of Buchholz relay.
12	To study the characteristics of the microprocessor based DMT/IDMT over current relay and determines the time current characteristics.
13	To study the characteristics of microcontroller based over current relay
14	To study microcontroller base over/ under voltage relay.
15	To study characteristics of electromechanical earth fault relay

Text Book

1. I.J.Nagrath, D.P.Kothari, Power System Engineering ,Tata Mc Graw-Hill publications, 2008
2. Ashfaq Husain, Electrical Power System,CBSPublication,5thEdition
3. C.L.Wadhwa,ElectricalPowerSystems,NewAgeInternationalPublisher,6thEdition
4. V.K.MehtaandRohitMehta,PrinciplesofPowerSystem,S.ChandPublication, 2008

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B.TECH. ELECTRICAL ENGINEERING**

Sem: IV	Total Hours Distribution per week		
Total Credit: 2	Lecture (L): 2 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BMD4T13	Elements of Electromagnetic	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Recognize and apply the knowledge of different co-ordinate systems.
2	Evaluate the physical quantities of electromagnetic fields in different media and apply Gauss law.
3	Describe static electric fields & explain steady magnetic fields
4	Understand Maxwell's equations in different forms and different media.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Review of Vector Analysis:			
Review of Scalars and vectors, Vector Algebra, Rectangular Co-ordinate System, Cylindrical Co-ordinate System, Spherical Co-ordinate System and transformation of Cartesian to Cylindrical, Cartesian to Spherical and vice versa.	7		1
Unit II: Coulomb's law, Electrical field intensity and electric flux density, Gauss's law, Divergence:			
Coulombs Law, Electric field intensity, field due to continuous volume charge distribution, field of point charge, field of line charge, field of sheet charge, Electric Flux density, Gauss's law and Applications of Gauss's law, the divergence theorem.	5		2
Unit III: Potential of charge system, poisson's and Laplace Equations & steady Magnetic Field			

Definition of potential difference and potential, the potential field of a point charge, the potential field of a system of charges, potential gradient, Poissons and Laplace Equation. Biot Savart's law, Ampere's Circuital law, Stoke's theorem, magnetic flux density, scalar and vector magnetic potentials	6		3
Unit IV: Boundary conditions, Maxwell's equation and wave propagation			
Magnetic boundary conditions, Faraday's law, Displacement current, Point form of Maxwell's equation, Integral form of Maxwell's equations, Wave propagation, Poynting vector, skin effect.	6		4

Suggested Textbooks &Reference Books

Textbooks:

1. W.H. Hayt , "Engineering Electromagnetic" ,TMH Publication 2006

Reference books:

1. Electromagnetic Engg. N.N.Rao V Edition ,Prentice Hall. 2005
2. Applied Electromagnetics, Fawwaz T.Ulaby, Prentice Hall. 1999
3. Krauss Electromagnetic Engg. IV Edition, Tata Mc Graw Hill. 2003
4. Electromagnetic Waves,Shevgaonkar Tata Mc Graw Hill 2002
5. Elements of Electromagnetics,Matthew, N. O. Sadiku, Oxford University publication, 6th edition, 2014.

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B.TECH. ELECTRICAL ENGINEERING**

Sem: IV	Total Hours Distribution per week		
Total Credit: 2	Lecture(L):2	Practical /Activity (T/A): NA.	
Subject Code	BOE4T02	Introduction to Power Systems	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives

1.	Understand basic electrical power concepts
2.	Learn about power generation methods
3.	Comprehend power transmission and distribution basics
4.	Explore renewable energy integration

Course Outcomes

After completion of syllabus, students would be able to

1.	Explain basic electrical power concepts
2.	Describe various power generation methods
3.	Understand power transmission and distribution basics
4.	Implement basic energy conservation measures

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit 1: Fundamentals of Electrical Power			
Basic electrical quantities (voltage, current, power), AC vs DC power systems Single-phase and three-phase systems, Power factor and its importance Simple circuit calculations, Units and conversions	6		1
Unit 2: Power Generation			
Conventional power plants: Thermal power plants, Hydroelectric power plants, Nuclear power plants (Block diagram description only) Renewable energy sources: Solar power, Wind power, Biomass Basic generator principles	6		2
Unit 3: Power Transmission & Distribution			
Transmission line components, Distribution system overview, Substations and transformers , Basic power loss concepts , Underground vs overhead lines Smart grid introduction	6		3
Unit 4: Energy Management			
Energy conservation principles, Power quality basics, Energy audit introduction , Load management , Energy efficiency measures , Smart metering	6		4

Textbooks

1. Power System Engineering, D.P. Kothari and I.J. Nagrath, Tata McGraw Hill Education, New Delhi
3rd Edition, Units 1, 2, 3.
2. Generation, Distribution and Utilization of Electrical Energy, C.L. Wadhwa, New Age International Publishers ,Latest Edition, Units 2, 3, 4
3. Basic Electrical Engineering, V.K. Mehta and Rohit Mehta, S. Chand Publishing ,Latest Edition

Reference Books

1. Power System Analysis, A. Chakrabarti and S. Halder, PHI Learning, Units 1, 3
2. Non-Conventional Energy Sources, G.D. Rai, Khanna Publishers, Unit 2 (Renewable Energy)
3. Electric Power Distribution System Engineering, Turan Gonen, CRC Press.

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
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FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING**

Sem: IV	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): NA	Practical /Activity (T/A): 4 Hr.	
Subject Code	BAE4P02	Technical Report Writing	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
50 Marks	50 Marks	50 Marks	---

This guideline provide the essential components for writing technical reports in engineering. Technical reports are formal documents that describe processes, progress, or results of technical or scientific research and investigations.

Structure of a Technical Report

Preliminary Pages	Main Body	End Matter
- Title Page - Report title - Author name(s) - Institution name - Date of submission - Course details -Abstract/Executive Summary - Brief overview of the entire report (200-300 words) - Table of Contents** - List of all sections and subsections with page numbers - List of figures and tables	Introduction, Background information, Problem statement, Objectives, Scope and limitations, Methodology overview - Literature Review - Summary of relevant research - Analysis of existing solutions - Theoretical framework - Critical evaluation of sources - Methodology - Detailed description of procedures , Equipment and materials used ,Experimental setup, Data collection methods Analysis techniques - Results and Discussion - Presentation of findings, - Data analysis, Interpretation of results, Comparison with literature ,Discussion of implications - Conclusions and Recommendations - Summary of key findings ,Practical implications, Suggestions for future work	- References - IEEE or other specified format - All cited sources listed - Appendices - Raw data - Detailed calculations - Supporting documents

Report Writing Process

Planning Phase	Research Phase	Writing Phase	Review Phase
- Understand the assignment requirements - Define scope and objectives - Create outline - Plan timeline	- Gather relevant literature - Collect data - Analyze findings - Document sources	- Draft main sections - Create figures and tables - Write introduction and conclusion - Prepare abstract	- Technical review - Grammar and style check - Format verification - Peer review if possible

Assessment Criteria

Reports are typically evaluated on:

1. Technical content and accuracy
2. Clarity and organization
3. Analysis and interpretation
4. Presentation and format
5. References and documentation

Practical Tips

1. Time Management
 - Start early
 - Break into manageable sections
 - Allow time for revision
 - Plan for unexpected delays
2. Documentation
 - Keep detailed notes
 - Document all sources
 - Save all raw data
 - Maintain version control
3. Professional Presentation
 - Consistent formatting
 - Clean, professional layout
 - High-quality graphics
 - Error-free content

**RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY,
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Sem: IV	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BHM4T02	Digital Economy	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
15 Marks	35 Marks	23 Marks	2 Hours

Course Outcomes	
After completion of syllabus, students would be able to	
1	Develop knowledge of economic theories and share market.
2	Identify key trends in Digital economy
3	Acquire knowledge of business strategies.
4	Analyze the problems of Small Scale Industries.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Concept of Economics			
Concept of Economics, Types of demand, Determinants of demand, Law of Demand, Elasticity of demand, Law of supply, Total utility and Marginal utility.	4		1
Unit II: Concept of Digital economy			
Concept of Digital economy, Components of digital economy: E-business, Infrastructure, E-commerce, Merits and Demerits of digital economy, Contribution of Digitalization in economic growth of country, Labour intensive verses Capital intensive industry.	8		2
Unit III: Concept of cost			
Concept of cost, Law of returns, Forward and Backward Integration, Merger and			3

Acquisition, Types of merger and acquisition, hostile takeover, advantages of merger and acquisition.	6		
Unit IV: Sickness in small business			
Sickness in small business, Major problems faced by SSIs, Technical consultancy organizations, Government Policies for Small Scale Enterprises, Tax holidays, Incentives to SSIs	6		4

Suggested Text Books &Reference Books

1. Modern Economic theory by K.K. Dewett, S Chand Publication, 2006.
2. The Digital Economy, by Tim Jordan, Polity publishers, 2020
3. Industrial economics and Entrepreneurship development by A.M. Sheikh, Nawaz Khan & M.A. Tonga, S chand Publication

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Sem: IV	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2 Hrs	Tutorial/Activity (T/A): 0 Hr.	
Subject Code	BVE4T02	Universal Human Values	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
15 Marks	35 Marks	23 Marks	2 Hours

Course Outcomes:	
After completing the course, students will be able to	
1	Understand the importance of human values and ethics for a harmonious life and society.
2	Develop clarity about relationships, happiness, and prosperity from a holistic perspective.
3	Apply universal human values in personal and professional life for ethical decision-making.
4	Evaluate the role of human values in sustainable development and social well-being.

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Introduction to Universal Human Values			
Introduction to Universal Human Values, Need for and Importance of Universal Human Values, Understanding Human Aspirations and Purpose of Life, Five Dimensions of Human Values: Individual, Family, Society, Nature, and Existence	4		1
Unit II: Understanding the Self			
Understanding the Self: "I" and the Body, Harmony of Thoughts, Behaviour, and Work ,Family as the Fundamental Unit of Society, Trust and Respect in Relationships , Ethical Values in Personal and Professional Life	8		2
Unit III: Society and Mutual Fulfilment			
Society and Mutual Fulfilment: Humanistic Education, Health, and Justice, Universal			

Order: Role of Ethics in Social Systems, Coexistence with Nature: Environmental Ethics and Sustainability , Holistic Perspective on Economic and Technological Growth	6		3
Unit IV: Holistic Development and Professional Ethics			
Holistic Development and Professional Ethics, Role of Human Values in Education, Science, and Technology, Ethical Dilemmas in Professional Life and Their Resolution, Corporate Social Responsibility (CSR) and Ethical Leadership	6		4

Suggested Text Books &Reference Books :

1. "Human Values and Professional Ethics" – R.R. Gaur, R. Sangal, G.P. Bagaria. Publisher: Excel Books
2. "Essence of Human Values" – A.N. Tripathi Publisher: New Age International
3. "Human Values" – Prof. A. Alavudeen, R. Kalil Rahman, and M. Jayakumaran Publisher: Laxmi Publications
4. "Professional Ethics and Human Values" – M. Govindarajan, S. Natarajan, V.S. Senthilkumar. Publisher: Prentice Hall India (PHI Learning)

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B.TECH. ELECTRICAL ENGINEERING**

Sem: IV	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): NA	Practical /Activity (T/A): 4 Hr.	
Subject Code	BVE4P02	Electrical Engineering Drawing	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
50 Marks	50 Marks	50 Marks	---

Sr. No	Title of Practical	Contents
1	Basic Electrical Components	-Resistors (fixed, variable, rheostats) . Capacitors (fixed, variable, electrolytic) , Inductors and transformers ,Power sources (AC/DC) ,Ground symbols and connecting terminals
2	Protection and Switching Devices	- Circuit breakers (MCB, MCCB, ACB) , Fuses and isolators, Switches (SPST, SPDT, DPST, DPDT) , Relays and contactors Push buttons and limit switches
3	Single-Line Diagrams	- Domestic electrical installation Industrial power distribution Transformer substation layout, Motor control center, Power factor improvement setup
4	Residential Installations	- Complete house wiring layout, distribution board connections, Earthing system layout, Light and fan circuit diagram, Energy meter installation
5	Industrial Installations	- Factory floor electrical layout ,Cable tray routing, Motor installation details Panel board arrangements Emergency lighting system

Fifth Semester Syllabus

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Sem: V	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hr	Practical /Activity (T/A): 0	
Subject Code	BEL5T13	Electrical Machines -II	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hrs

Course Objectives

1	To study Speed control and braking methods for Induction Motor.
2	To understand voltage regulation and parallel operation of alternators.
3	To evaluate performance of synchronous machines.
4	To analyze transient behavior of synchronous machines.
5	To apply special machines used in EVs, renewable energy, and automation.

Course Outcomes

After completion of syllabus, students would be able to

1	Evaluate Speed control and braking methods of induction motors used in industrial Drives.
2	Understand voltage regulation and parallel operation of alternators along with synchronization.
3	Analyze the operating characteristics and power flow performance of synchronous generators connected to power systems.
4	Understand transient behavior of synchronous machines.
5	Assess special machines used in electric vehicles, robotics, and renewable energy systems.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Induction Motor			
Dynamic model of induction machine in stationary reference frame, d-q axis model of induction machine Speed Control: Variable Frequency control, Stator voltage control, Pole changing methods, cascaded connection, rotor resistance control. Electric braking methods : Dynamic or rheostatic, plugging and regenerative braking.	9	0	1

Unit II: Voltage regulation & Parallel operation of Alternator			
Equivalent circuit and phasor diagrams of a synchronous generator, Voltage regulation by Zero power factor (ZPF) method, Potier triangle. Parallel operation of alternators: Reasons and necessary conditions, Synchronization, Synchronizing procedure.	9	0	2
Unit III: Synchronous Machine Performance			
Machine equations in rotor reference frame, d-q axis model of synchronous machine, Two reaction theory of salient pole synchronous machine, Power flow in synchronous motor, phasor diagram of salient pole rotor machine, slip test, expression of synchronizing power & synchronizing torque, Hunting & prevention, comparison of synchronous motor & Induction Motor, applications.	9	0	3
Unit IV: Transient behavior of synchronous machines			
Short circuit ratio, sequence components, sudden three phase short circuit, constant flux linkage theorem, Transient, sub-transient reactance's & time constants, equivalent circuit diagram, determination of negative and zero sequence reactance.	9	0	4
Unit V: Special Machines			
Permanent Magnet Synchronous Motor (PMSM), Switched Reluctance Motor (SRM), Stepper Motor, BLDC motors used in electric vehicle propulsion systems, Servo motors used in automation, Applications in robotics and renewable energy systems.	9	0	5

Textbooks :

1. P. S. Bimbhra – Electrical Machinery
2. I. J. Nagrath & D. P. Kothari – Electrical Machines
3. Stephen Chapman – Electric Machinery Fundamentals
4. Fitzgerald & Kingsley – Electric Machinery
5. Janardanan, E.G. - Special Electrical Machines
6. Keerthi Kumar N et. al. –E - Vehicle Renewable Energy
7. B.L.Thareja , Electrical Machines , Volume -2 ,

Reference Books :

1. "Analysis of Electric Machinery and Drive Systems" by P.C. Krause, O. Wasynczuk, and S.D. Sudhoff.
2. "Brushless Permanent Magnet and Reluctance Motor Drives" by T.J.E. Miller
3. "Generalized Theory of Electrical Machines" by P.S. Bimbhra

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Sem: V	Total Hours Distribution per week		
Total Credit: 1	Practical /Activity (T/A): 2 Hr		
Subject Code	BEL5P13	Electrical Machines -II Lab	
Examination Scheme			
Practical Internal Marks:	Practical University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	---

Practical Course Objectives	
1	Conduct experiments on AC machines to understand their operating characteristics and performance.
2	Apply control techniques for speed regulation, braking, and excitation of induction and synchronous machines.
3	Perform testing, measurement, and parameter determination of alternators and synchronous machines.
4	Explore and implement modern electrical machine technologies such as BLDC and Switched Reluctance Motors.
5	Use simulation tools (MATLAB/Simulink, SRM software) to model, analyze, and visualize machine performance.

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Demonstrate speed control, braking, and excitation control of three-phase induction and synchronous machines.
2	Conduct open-circuit, short-circuit, loading, and synchronization tests on alternators and evaluate their performance.
3	Determine machine parameters including direct-axis, quadrature-axis, sub-transient reactance, voltage regulation, and SCR.
4	Operate and study BLDC motors and modern electrical machines with electronic control.
5	Simulate electrical machines using d–q axis modeling, dynamic models, and SRM simulations to predict performance under various conditions.

A minimum of eight experiments, aligned with the syllabus of Electrical Machines – II, shall be conducted, subject to the availability of laboratory equipment and relevant software resources. Additionally, at least one experiment shall be carried out using a virtual laboratory platform. The following experiments are recommended:

Pr. No.	List of Practical
1	To perform speed control of a three-phase induction motor by stator voltage control.
2	To perform electric braking of a three-phase induction motor and to study dynamic (rheostatic) braking and plugging methods.
3	To perform open circuit test (OCC) and short circuit test (SCC) on a three-phase alternator and to obtain the characteristic curves.
4	To determine the voltage regulation of an alternator by Zero Power Factor (ZPF) method.
5	To determine the voltage regulation of an alternator using the Potier triangle method.
6	To perform direct loading test on a three-phase alternator and to obtain the characteristic curves
7	To perform synchronization of an alternator with the infinite bus using lamp method and synchroscope.
8	To perform parallel operation of two alternators and to study load sharing between them.
9	To perform slip test on a salient pole synchronous machine and to determine direct axis reactance (X_d) and quadrature axis reactance (X_q).
10	To perform the sudden three-phase short-circuit test on a synchronous machine in order to determine its direct-axis subtransient reactance (X_d'') and quadrature-axis subtransient reactance (X_q'').
11	To perform the V-curve and inverted V-curve test on a synchronous motor and to study the effect of field excitation on armature current.
12	To determine the short circuit ratio (SCR) of an alternator using open circuit and short circuit characteristics.
13	To determine the negative sequence and zero sequence reactance of a synchronous machine.
14	To study the operation and speed control of a Brushless DC (BLDC) motor.
Simulation based:	
1	To simulate the operation of a Switched Reluctance Motor (SRM) and to study its torque characteristics.
2	To simulate the dynamic model of an induction machine in stationary reference frame using MATLAB/Simulink and to study the machine performance.
3	To simulate the d-q axis model of an induction machine and to observe the variation of torque, flux and currents under dynamic conditions.
4	To simulate sudden three-phase short circuit on a synchronous generator and to study transient and sub-transient current response.
5	To simulate the d-q axis model of a synchronous machine in rotor reference frame and to study the machine performance.

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B.TECH. ELECTRICAL ENGINEERING

Sem: V	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2 Hr	Practical /Activity (T/A): 0	
Subject Code	BEL5T14	Electrical Power Utilization	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	2 Hrs

Course Objectives

1	Introduction of various methods of electric heating.
2	Understand various methods of electric welding.
3	Understand the principles and design considerations of illumination systems.
4	Study the working principles and applications of refrigeration and air conditioning systems for domestic and industrial purpose.

Course Outcomes

After completion of syllabus, students would be able to	
1	CO1: Explain the principles, methods, and applications of electric heating systems.
2	CO2: Describe different electric welding techniques.
3	CO3: Analyze the concepts of illumination system .
4	CO4: Discuss the different refrigeration and air-conditioning systems .

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Electric Heating:			
Types and methods of electrical heating, advantages of electrically produced heat, types & application of electric heating equipment's, Resistance Ovens: General constructions, design of heating elements, efficiency & losses. Induction heating: Core type & core less induction furnace, indirect induction oven. Dielectric heating: Principle and application. Arc furnace : Direct & indirect arc furnace	7	0	1
Unit II: Electric Welding			
Introduction, advantages and classification of electric welding processes.			2

Resistance welding: Butt welding, Spot welding, Projection welding, Electric arc welding: Carbon arc welding, metal arc welding. Special welding: Stainless Steel welding, Ultrasonic welding, laser beam welding	8	0	
Unit III: Illumination			
Nature of light, terms used in illumination, solid angle, laws of illumination, polar curves, Color Rendering Index (CRI), Design of illumination systems, types of lighting energy saving in lighting systems.	7	0	3
Unit IV: Refrigeration and Air Conditioning			
Refrigeration: refrigeration cycle, refrigeration systems (Vapor compression, vapor absorption), domestic refrigerator, drinking water cooler, desert air cooler ,Air conditioning: Factors involved in air conditioning, comfort air conditioning, industrial air conditioning, summer / winter air conditioning systems	8	0	4

Textbooks:

1. Fundamental of Electrical Drives by G K Dubey
2. Utilization of Electrical Power by R K Rajput
3. Utilization of Electrical Energy and Traction by J.B. Gupta
4. Utilization of Electrical Energy by S.K. Sahdev.

Reference Book:

1. Utilization of Electrical Energy by E. Openshaw Taylor (Orient Longman)
2. Electric Energy Utilization and Conservation by S. C. Tripathy.

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Sem: V	Total Hours Distribution per week		
Total Credit: 1	Practical /Activity (T/A): 2 Hr		
Subject Code	BEL5P14	Electrical Workshop Lab.	
Examination Scheme			
Practical Internal Marks:	Practical University Marks:	Minimum Passing Marks:	Examination Duration:
50 Marks	50 Marks	50 Marks	---

Practical Course Objectives	
1	Design, implement, and test domestic and small residential electrical circuits with proper protective devices.
2	Test and troubleshoot electrical appliances and circuits, identifying faults and ensuring safe operation.
3	Apply electronics concepts to design and implement control circuits, including DC motor controllers, LDR-based circuits, and voltage protection systems.
4	Develop skills in soldering, de-soldering, and PCB assembly, ensuring reliable electrical/electronic connections.
5	Integrate modern devices such as LED lighting, electronic regulators, inverters, and UPS systems into practical installations.

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Design, assemble, and test domestic lighting and power circuits with multiple loads, MCBs, fuses, and earthing.
2	Measure insulation resistance of cables, motors, and appliances , and interpret results for safety compliance,
3	Perform soldering, de-soldering, and PCB assembly, ensuring functional and reliable electronic connections.
4	Design, build, and test control circuits such as DC motor controllers, light-sensitive relays, and voltage stabilizers.
5	Develop the ability to analyze, evaluate, and troubleshoot electrical and electronic systems in practical settings.

A minimum of eight experiments shall be conducted, subject to the availability of laboratory equipment and relevant software resources. The following experiments are recommended:

Pr. No.	List of Practical
1	Test a domestic lighting circuit with multiple loads and protective devices (MCB, fuse).
2	Verify operation and troubleshoot faults of Staircase wiring with two-way and intermediate switches
3	Test a fluorescent lamp or LED lighting circuit with proper control and dimming.
4	Test ceiling fan wiring with electronic regulator; measure and analyze current drawn at different speeds.
5	test a small residential power circuit with socket outlets and protective earthing.
6	Measurement of insulation resistance of cables, motors, and appliances using a Megger.
7	Testing and comparison of protective devices: fuses, MCBs, ELCBs, and RCCBs under fault conditions.
8	Troubleshooting a faulty domestic circuit: short circuits, open circuits, and loose connections.
9	Testing and fault analysis of household appliances such as fan, heater, iron, and refrigerator.
10	Soldering, de-soldering, and assembling of PCB-based simple circuits; testing of joints.
11	Build and test a DC motor controller using relays and transistors; verify speed control.
12	Design and implement a basic light-sensitive circuit using LDR, transistor, and relay.
13	Build a simple voltage stabilizer/protection circuit and test with variable loads.
14	Implement and test a small inverter or UPS circuit for low-power loads
15	Design of small single phase shell type transforemr

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Sem: V	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hr	Practical /Activity (T/A): 0	
Subject Code	BEL5T15	Power Systems-II	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3Hrs

Course Objectives

1	To develop the ability to understanding power system concepts like analysis of symmetrical and unsymmetrical faults using symmetrical components as a tool.
2	Knowledge of power system stability
3	Appreciation of economics, control & management of power system
4	Optimize the combination of available generating units to minimize total costs, including start-up and shut-down costs.
5	Maintain grid stability by balancing Active Power for Frequency Control, managing Reactive Power for Voltage Control, and adjusting Network Parameters for Power Flow Control.

Course Outcomes

After completion of syllabus, students would be able to

1	Apply symmetrical components concepts in fault analysis
2	Evaluate fault currents for different types of faults
3	Appreciate concepts of power system stability.
4	Understand economic operation of power system.
5	Understand methods to control the voltage, frequency and power flow.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Symmetrical Component transformation:			
Three phase power in unbalanced circuit in terms of symmetrical component. Sequence impedances of Generator. Transformer Transmission line & Passive loads. Phase shift in star - delta three phase transformer.	8		1

Unit II: Symmetrical & Unsymmetrical fault analysis:			
Symmetrical fault analysis without & with pre fault load current. L-G, L-L-G, L-L, open conductor faults analysis using symmetrical components. Selection of Circuit Breakers ratings, current limiting reactors.	11		2
Unit III: Stability of Power System			
Steady state, Dynamic and Transient stability definition. Dynamics of synchronous machine, swing equation, swing equation for machines swinging coherently and Non-Coherently. Power angle equation. Steady state stability studies. Transient stability studies: - Swing curve, Equal Area criterion for transient stability. Application of equal area criterion for different disturbances. Solution of swing equation by point by point method. Methods of improving transient stability.	10		3
Unit IV: Economic operation of power system:			
Introduction, Distribution of load between units Within the plant Optimum generation scheduling considering transmission losses. Representation of transmission loss using loss formula coefficient. Derivation of loss formula coefficient, simulation of co-ordination equation on digital computer.	8		4
Unit V: Power System Dynamics			
Turbines and speed governors, frequency dependence of loads, Droop control and power sharing. Automatic generation control, Generation and absorption of reactive power by various components of power system. Excitation system control in synchronous generators, automatic voltage regulators.	8		5

Suggested Text books & Reference Books

Text Books

1. W.D. Stevenson Jr., Elements of power system analysis, McGraw-Hill publications, 3rd Edition
2. O. I. Elgerd, —Electric Energy Systems Theory, McGraw Hill Education, 1995.
3. A. R. Bergen and V. Vittal, —Power System Analysis, Pearson Education Inc., 1999.
4. D. P. Kothari and I. J. Nagrath, —Modern Power System Analysis, McGraw Hill Education, 2003.
5. C. L. Wadhwa- Electrical Power System, New Academic Science Limited, 2012

Reference Books

1. Hadi Saadat, Power System Analysis ,TMH , 2002
2. J. Grainger and W. D. Stevenson, —Power System Analysis, McGraw Hill Education, 1994.
3. Prabha Kundur, Power System Analysis and Control , TMH, 2008
4. B. M. Weedy, B. J. Cory, N. Jenkins, J. Ekanayake and G. Strbac, —Electric Power Systems, Wiley, 2012.

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Sem: V	Total Hours Distribution per week		
Total Credit : 4	Lecture (L): 3 Hr	Practical /Activity (T/A): 1Hr	
Subject Code	BEL5T16A	PLC and SCADA	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hr

Course Objectives

1	To have the basic concept, components and architecture of PLC used in automation systems.
2	To select appropriate PLC modules and hardware components as per industrial application requirements.
3	To implement and develop ladder logic programs for various automation applications.
4	To understand the basic concepts, architecture, and displays of SCADA systems.
5	To design and develop SCADA-based monitoring and control systems for different industrial applications.

Course Outcomes

After completion of syllabus, students would be able to	
1	Identify and understand components of PLCs for Automation
2	Select appropriate module as per application.
3	Develop PLC Programming for given application.
4	Understand SCADA System.
5	Develop SCADA system for various applications.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Introduction to PLC			
Need and tools of Automation, Evolution of PLC, Architecture- PLC Block diagram and working, Selection of PLC, Types of PLCs, Advantage,	8	2	1

limitations and applications of PLCs, Networking of PLCs.			
Unit II: PLC Hardware			
Input and Output Modules for PLC-working, description, wiring details, specifications, interfacing; Instruction sets for given operation, Ladder logics for some applications.	9	4	2
Unit III: PLC Programming and Applications			
Programming Languages for PLCs, PLC programming standard, Relay type Instructions-Timer, Counter, Arithmetic operation, Data handling instructions. PLC based applications as motor control, traffic light, etc.	10	3	3
Unit IV: SCADA Systems and Human Machine Interface			
Introduction to SCADA systems and architecture, Components of SCADA: Remote Terminal Units (RTU) PLC Communication system SCADA server Human Machine Interface (HMI) Functions of SCADA: Data acquisition Supervisory control Alarm management Historical data logging Trend analysis SCADA software features Design of HMI screens Applications in: Power systems monitoring Water treatment plants Oil and gas industries	9	3	4
Unit V: Industrial Communication and Advanced Automation			
Industrial communication networks Modbus Profibus DeviceNet Industrial Ethernet PLC networking concepts PLC-SCADA communication OPC (OLE for Process Control) Cybersecurity in industrial automation Introduction to: Distributed Control Systems (DCS) Industrial Internet of Things (IIoT) Industry 4.0 concepts	9	3	5

Suggested Text books & Reference Books

Text Books

1. Programmable Logic Controllers – W. Bolton, 6 th Edition, 2015, Newnes, Elsevier Ltd.
2. Practical SCADA for industry – Bailey David, Wright Edwin, 1 st Edition, 2003, Newnes.

Reference Books

1. Programmable Logic controllers and Industrial Automation: Madhuchhanda Mitra, Samarjit Sen Gupta, Penram International Publishing India Pvt. Ltd
2. PLC and SCADA System: Karambir, Rajiv Kumar, Dreambook Publishing, 2023.
3. PLC Programming for Industrial Automation: Kevin Collins, Exposure Publishing, United Kingdom.
4. Programmable Logic Controllers and Ladder Logic: Greg P. Zimmerman, South Dakota School of Mines and Technology, April 2008.
5. Supervisory Control and Data Acquisition: S.A. Boyar, ISA Publication.
6. Programmable Logic controllers: V.R. Jadhav, Khanna Publications.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: V	Total Hours Distribution per week		
Total Credit: 4	Lecture (L): 3 Hr	Practical /Activity (T/A): 1 Hr	
Subject Code	BEL5T16B	Electrical Machine Design	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hr

Course Objectives

1	Understand materials, insulation, cooling methods, and ratings used in electrical machines.
2	Learn the design principles of single-phase and three-phase transformers.
3	Analyze performance characteristics of transformers.
4	Understand the design of stator and rotor of three-phase induction motors.
5	Learn the design principles of three-phase alternators.

Course Outcomes

After completion of syllabus, students would be able to	
1	Select suitable materials, insulation, and cooling methods for electrical machines.
2	Design single-phase and three-phase transformers.
3	Evaluate transformer performance parameters.
4	Design stator and rotor of three-phase induction motors.
5	Design three-phase alternators.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Review of Material used in Construction of Electrical Machines			
Classification of insulating materials depending upon permissible temperature rise, properties of transformer oil. Standard specification, C.M.R. and short-time rating of machines. Heating and cooling characteristics.	7	2	1
Unit II: Design of 1-phase and 3-phase core-type transformer			

Classification of transformers based on construction and service conditions, Output equation, overall dimensions of 1-phase and 3-phase core type transformer, need of stepped core cross-section, selection of flux density and current density, design of the transformer for minimum cost, minimum weight, and minimum losses.	10	4	2
Unit III: Performance characteristics of Transformer			
Type of windings used in transformer, Calculations of per unit leakage reactance and regulation for core type transformer, No load current calculations, Introduction to cooling Methods.	9	3	3
Unit IV: Stator Core & Rotor Design of 3-phase Induction Motor			
Stator Core Design: Output equation of 3-phase Induction motor, selection of specific magnetic loading & specific electric loading, Selection of number and type of stator slots, overall dimensions of the stator core. Rotor Design: Classification of 3-phase Induction motor based on rotor construction, selection of length of air gap and rotor slots, calculations of overall dimension and rotor speed of 3-phase squirrel cage rotor of 3-phase Induction motor.	10	4	4
Unit V: Design of 3-phase alternator			
Classification of the alternators, peripheral speed and runaway speed, pitch factor and distribution factor. Effect of SCR on the performance of the alternator, overall dimensions of 3 phase alternator.	9	2	5

Text Books:

1. Electrical Machine Design by A.K. Sawhney, Dhanpatrai and Sons, Delhi.
2. Electrical Machine Design by Balbir Singh Brite students Publication, Pune.
3. Principles of Electrical Machine Design by R. K. Agarwal, S. K. Katariya & Sons, Delhi.

Reference Books:

- 1 Performance and Design of A.C. Machines by M.G. Say, CBS Publishers.
2. Design and Testing of Electrical Machines by M.V. Deshpande, PHI publishers.
3. Electrical Machine Design by V. Rjini, V.S. Nagarjan, Pearson Publishers.

Data Books:

1. Electrical Machine Design data book, A. Shanmugsundaram, G. Gangadharan, P. Palani, New Age International Publishers.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: V	Total Hours Distribution per week		
Total Credit : 4	Lecture (L): 3 Hr	Practical /Activity (T/A): 1Hr	
Subject Code	BEL5T16C	Digital Electronics Circuits	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hours

Course Objectives

1	To provide a strong foundation in number systems and codes used in digital systems.
2	To enable students to understand and apply Boolean algebra and logic minimization techniques for simplifying digital circuits.
3	To develop the ability to analyze and design combinational digital circuits such as multiplexers, decoders, adders, encoders, and display drivers.
4	To impart knowledge of sequential circuit design using flip-flops, shift registers, and counters for real-world digital applications
5	To familiarize students with the working principles of DAC and ADC converters, semiconductor memory organization, and programmable logic devices.

Course Outcomes

After completion of syllabus, students would be able to

1	Convert number system, comprehend logic gates and logic families.
2	Apply Boolean algebra theorems and Karnaugh map techniques to simplify logic functions.
3	Design and analyze combinational digital circuits.
4	Analyze and design sequential circuits.
5	Evaluate DAC and ADC conversion techniques and understand the memory organization.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Fundamentals of Digital Systems and Logic Families			
Number systems-binary, signed binary, binary arithmetic, one's and two's complements arithmetic, octal and hexadecimal number system, codes, Basic digital circuits using basic gates, NAND and NOR operations, Exclusive – OR	10	3	1

and Exclusive NOR operations, Gate ICs, Digital logic families- RTL, TTL and CMOS logic, interfacing CMOS and TTL.			
Unit II: Logic Function and minimization			
Boolean algebra, minimization of logical functions. Standard representation for logic functions, K-map representation (up to 4 variables), and simplification of logic functions using K-map, Don't care conditions	7	3	2
Unit III: Combinational Digital Circuits			
Multiplexer, De-Multiplexer, Decoders, Adders, Subtractors, code converters, priority Encoders, decoders/drivers for display devices – Seven Segment display	9	3	3
Unit IV: Sequential circuits and systems			
A 1-bit memory, Bi-stable latch, Clocked SR flip flop, J- K flip flop, T and D types flip-flops, excitation table of flip flop, conversion of flip flops, applications of flip flops: Shift registers and applications of shift registers (SIPO, PIPO); Counters (Upto 4 bit)- ripple (Asynchronous) counters, synchronous counters, ring counter, applications of counters- Traffic light control	9	3	4
Unit V: Converters and Semiconductor memories			
Digital to analog converters: weighted resistor/converter, R-2R Ladder D/A converter, examples of D/A converter ICs, Analog to digital converters: successive approximation A/D converter, counting A/D converter, dual slope A/D converter, A/D converter ICs. Memory organization and operation, expanding memory size, classification and characteristics of memories, Programmable logic devices (PLD): Programmable logic array, Programmable array logic, ROM/PROM.	10	3	5

Suggested Text books & Reference Books

Text Books

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.
3. H.Taub, "Digital Integrated Electronics" McGraw Hill
4. A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016.

Reference Book:

1. Herbert Taub, Donald L Schilling "Digital Integrated Electronics", McGraw Hill, 1977
2. Thomas C Bartee, "Digital Computer Fundamentals", McGraw Hill, 1985.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: V	Total Hours Distribution per week		
Total Credit: 3	Lecture (L): 3 Hr	Practical /Activity (T/A): 0 Hr	
Subject Code	BMD5T17	Programming Techniques & Simulation	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	---

Course Objectives

The objective of this course is to introduce basic **C, C++, and MATLAB programming** and simulation tools, enabling students to solve engineering problems, perform numerical computations, visualize data, and model basic electrical systems using MATLAB and Simulink

Course Outcomes

After completion of syllabus, students would be able to

1	Understand the structure and syntax of C programs and develop basic programs using data types, operators, and control statements
2	Explain and apply object-oriented programming concepts in C++ , including classes, objects, inheritance, polymorphism, encapsulation, and data abstraction.
3	Write, save, and execute basic MATLAB programs for data input/output, formatting results, and simple data visualization
4	Use conditional and iterative statements in MATLAB to solve basic computational and logical problems
5	Perform matrix operations , generate basic plots and waveforms, and understand the fundamentals of MATLAB toolboxes and Simulink for simple simulation tasks

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO

Unit-I			
Structure of C program, Data types, Variables, Input/output statements, Storage class, operators, Program control statements, Basic C Programs.	8	0	1
Unit II			
Introduction to C++ concepts, objects and classes in C++, difference between object oriented programming and procedure oriented programming, Data Abstraction, Polymorphism, Data hiding, Encapsulation, Multiple Inheritance.	10	0	2
Unit III			
Introduction to MATLAB programming concepts, MATLAB program structure and syntax, Writing, saving, and executing simple scripts (M-files), Importing and exporting data files, Displaying output and formatted results, Using graphics tools to display basic data.	10	0	3
Unit IV			
Conditional statements: if statement, if-else statement, Iterative statements: for loop, while loop, Program development using control structures, Simple problem-solving using MATLAB programs	8	0	4
Unit V			
Matrix operations: Transpose of a matrix, Determinant of a matrix, Inverse of a matrix Plotting of graphs using MATLAB: Basic 2D plots, Generation of simple waveforms Programming using MATLAB functions, Introduction to MATLAB Toolboxes: SimPower System Toolbox, Introduction to Simulink and basic simulation concepts	9	0	5

Text Books:

1. Venugopal, "Pascal & C Programming", Tata McGraw-Hill Education Pvt. Ltd.
2. Y. Kanetkar, "Let Us C", BPB Publications, 8th Edition.
3. Balguru Swami, "Computer Programming in C", Tata McGraw-Hill Education Pvt. Ltd.
4. Ashwin Pajankar, Arduino Made Simple with Interactive Projects, BPB Publications, 1st Edition.

Reference Books:

1. B. W. Kernighan and D. M. Ritchie, "The C Programming Language", Pearson Education, 2nd Edition.
2. Stormy Attaway, "MATLAB: A Practical Introduction to Programming and Problem Solving", Elsevier.
3. Duane Hanselman and Bruce Littlefield, "Mastering MATLAB 7", Pearson Education.
4. Rohan Barnwal, Arduino Programming Projects, BPB Publications, 1st Edition.
5. Jack Purdum, Beginning C for Arduino, Apress / Springer, 2nd Edition.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: V	Total Hours Distribution per week		
Total Credit: 1	Practical /Activity (T/A): 2 Hr		
Subject Code	BMD5P17	Programming Techniques & Simulation Lab	
Examination Scheme			
Practical Internal Marks:	Practical University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	---

10 EXPERIMENTS BASED ON ABOVE THEORY SYLLABUS.

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FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: V	Total Hours Distribution per week		
Total Credit : 2	Lecture (L): 2 Hr	Practical /Activity (T/A): 0	
Subject Code	BOE5T03	Power Plant Engineering	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 hrs

Course Objectives

1	Understand various energy sources, power tariff types, load distribution parameters and load curves.
2	Explain the working, layout and subsystems of coal-based thermal power plants.
3	Describe hydrology concepts, components and classification of hydroelectric power plants.
4	Understand nuclear power plant fundamentals, types of reactors and waste disposal methods.

Course Outcomes

After completion of syllabus, students would be able to

1	Compare conventional and non-conventional energy sources and explain tariff structures, load distribution parameters.
2	Explain the working of coal-based thermal power plants including major components.
3	Interpret flow duration and power duration curves and describe layout and components of hydroelectric power plants.
4	Explain the construction and operation of different nuclear reactors (BWR, PWR, CANDU, Breeder, Gas-cooled) and waste disposal methods.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Various Sources of Energy, Tariff & Load Distribution Parameters			
Resources and Development of Power in India, Introduction to the Sources of Energy (Solar, Wind, Hydroelectric, Nuclear, Fossil Fuels, Geothermal, Tidal) Its advantages & Disadvantage, its comparison,	8		1

Power tariff & its types, Load distribution parameters, load curve, load duration curve, base load and peak load plants, types of loads with its characteristics, numerical based on load distribution parameter & load curve.			
Unit II: Coal Based Thermal Power Plants*			
Rankine cycle - improvisations, Choices of site selection, Layout of modern coal power plant, Its Major Equipment's (Boilers, Turbines, Condensers, Economizer, Air-preheater etc). Subsystems of thermal power plants– Fuel and ash handling plant, Draught system, Feed water treatment. Waste Disposal Options for Coal Power Plants.	7		2
Unit III: Hydro-Electric Power Plants: *			
Hydrology, Flow duration curve, power duration curve, mass curve, typical layout of HPP, its advantages and disadvantages, and various components, its Classification, Pumped storage hydroelectric power plant & their utility.	8		3
Unit IV: Nuclear Power Plants*			
Basics of Nuclear Engineering, Layout and subsystems of Nuclear Power Plants, Working of NPP, Introduction to Nuclear Reactors (Purposeonly) ,waste disposal options for nuclear plants. Scenario of Nuclear power plant in India & globally.	7		4

***No Numerical Expected**

Suggested Text books & Reference Books

1. Nag. P.K., "Power Plant Engineering", Third Edition, Tata McGraw – Hill Publishing Company Ltd.,2008.
2. Generation of Electrical energy By B.R. GuptaS. Chand publication. 1 November 2017
3. 2. A Textbook of Power Plant Engineering by R.K. Rajput | 1 January 2016.
4. An Introduction to Power Plant Technology / G.D. Rai, Khanna Publishers, 1 January 1987

REFERENCES:

1. El-Wakil. M.M., "Power Plant Technology", Tata McGraw – Hill Publishing Company Ltd., 2010.
2. Godfrey Boyle, "Renewable energy", Open University, Oxford University Press in association with the Open University, 2004.
3. Thomas C. Elliott, Kao Chen and Robert C. Swanekamp, "Power Plant Engineering", SecondEdition, Standard Handbook of McGraw – Hill, 1998.
4. Power Plant Engineering by B. Vijaya Ramnath C. Elanchezhian, L. Saravanakumar | 1 November 2019
5. Power Plant Engineering, As per AICTE: Theory and Practice by Dipak Kumar Mandal, Somnath Chakrabarti, et al. | 1 January 2019

Sixth Semester Syllabus

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit: 3	Lecture (L): 3 Hr	Practical /Activity (T/A): 0	
Subject Code	BEL6T18	Power Electronics	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hr

Course Objectives

1	Review power semiconductor devices and their protection methods.
2	Analyze controlled rectifier circuits.
3	Design DC–DC converters and their control techniques.
4	Describe the operation of DC–AC inverters and PWM methods.
5	Explore AC controllers, cycloconverters, and practical applications.

Course Outcomes

After completion of syllabus, students would be able to

1	Analyze the operation and characteristics of power semiconductor devices.
2	Design and evaluate single and three-phase rectifiers.
3	Design DC–DC converter circuits for power control applications.
4	Analyze inverter circuits and apply PWM techniques.
5	Apply power electronics concepts to renewable energy, EVs, and power system

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-I: Introduction to Power Semiconductor Devices			
SCR: SCR, its characteristics (V-I, Dynamic characteristics), GATE Characteristics, Two Transistor Analogy , UJT relaxation oscillator, Series & Parallel connections (only introduction).	10		1

Static controllable switches: Structure & characteristics of MOSFET, IGBT, TRIAC Commutation techniques of SCR: Natural commutation (only working), Forced Commutation A, B, C, D, E & F (only working).			
Unit II: Phase-Controlled Rectifiers			
Single-phase controlled rectifiers: half-controlled and fully controlled bridge rectifiers with R and RL loads. Introduction to Semi converter. Three-phase rectifiers: Three-phase half-wave and full-wave controlled bridge rectifiers with R & RL loads. Effect of source inductance and freewheeling diode. Dual converters: circulating current type & non-circulating current type.	10		2
Unit III: DC–DC Converters (Choppers)			
Buck (Step Down), Boost (Step Up), Buck-Boost, Two Quadrant & Four Quadrant operation of Chopper, Output Voltage Control Techniques: Current limit control, Time ratio control.	8		3
Unit IV: DC–AC Inverters			
1-phase bridge inverter & 3 ϕ bridge inverters (120° & 180° Modes), output voltage control, Harmonic reduction by PWM Techniques (SPWM, MPWM, SinPWM), Introduction to current source Inverter, Comparison of VSI & CSI.	9		4
Unit 5: AC Voltage Controllers, Cycloconverters, and Applications			
Single-phase AC voltage controllers with R and RL loads, AC phase control using TRIAC, Principle and operation of cycloconverters (single-phase) Applications of power electronics: Renewable energy systems, Electric vehicles, battery charging)	8		5

Suggested Textbooks & Reference Books

Text Books

- 1) Power Electronics circuits Devices, and Applications by M. H. Rashid, Prentice Hall India
- 2) Power Electronics, by M. D. Singh & Khanchandani, Tata McGraw-Hill
- 3) Power Electronics by P. C. Sen.
- 4) Thyristors and their Applications by G. K. Dubey and, Doralda Joshi and Sinha, New Age

Reference Books

- 1) Power Electronics, Ned Mohan, Tora M. Udeland, William P. Riobbins, John Wiley & Sons

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit: 1	Practical: 2 Hr		
Subject Code	BEL6P18	Power Electronics Lab	
Examination Scheme			
Practical Internal Marks:	Practical University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	---

Practical Course Objectives	
1	To study the characteristics and operating principles of power semiconductor devices such as SCR and UJT.
2	To understand triggering methods and control techniques used in SCR-based circuits.
3	To analyze the operation of controlled rectifiers and AC voltage regulators with different loads.
4	To study the working principles of DC–DC converters and their output control
5	To analyze single-phase and three-phase inverter circuits and observe output waveforms.

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Analyze the V–I characteristics of SCR and determine latching and holding currents.
2	Design and test triggering circuits for SCR-based power electronic applications.
3	Evaluate the performance of controlled rectifiers and AC voltage regulators with R and L loads
4	Analyze the operation and output characteristics of DC–DC converters
5	Implement inverter circuits and analyze voltage and current waveforms using PWM control techniques.

List of Practical:

Pr. No.	List of Practical
1	To study the V–I characteristics of SCR
2	To determine the latching current and holding current of SCR
3	To study the UJT triggering circuit for half-wave and full-wave control of SCR.
4	To study a single-phase half-wave-controlled rectifier with (i) Resistive load, (ii) Inductive load with and without freewheeling diode.
5	To study a single-phase full-wave-controlled rectifier with (i) Resistive load, (ii) Inductive load with and without freewheeling diode.
6	To study a single-phase AC voltage regulator with resistive and inductive loads.
7	To study the operation of buck converter and variation of output voltage with duty cycle.
8	To study the operation of boost converter and variation of output voltage with duty cycle.
9	To study the operation of buck-boost converter and variation of output voltage with duty cycle.
10	To study the operation of a single-phase cycloconverter.

11	To implement a single-phase full-bridge inverter and analyze output voltage and current waveforms.
12	To implement PWM control in a single-phase inverter and observe the effect on output voltage
13	To implement a three-phase bridge inverter and observe phase and line voltages.

* Any Eight practical's from the above-mentioned practical's can be performed

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FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit : 4	Lecture (L): 3 Hr	Practical /Activity (T/A): 1	
Subject Code	BEL6T19	Control System	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hr

Course Objectives

1	To study modeling and transfer function of linear time-invariant system
2	To understand the stability, time domain specifications and tools
3	Determine system stability using Routh-Hurwitz criterion, Root Locus, Bode plots, and Polar plot.
4	To study frequency domain analysis of linear system
5	An introduction to state space approach and to understand the theory state transition matrix

Course Outcomes

After completion of syllabus, students would be able to

1	Model the linear system and study the control system components specifications through classical approach
2	Comprehend the time response and it's specifications and different controllers.
3	Analyze the absolute stability and relative stability through Routh's stability and root locus technique.
4	Evaluate frequency-dependent system behavior and relative stability margins using Bode, Polar, and Nyquist plots.
5	Understand the concept of state variable approach

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Introduction to Control System:			
Need of control system, Open loop and Closed loop control, Significance of actuators and sensors, Control system component's (DC/AC servomotor, potentiometer, synchro), Mathematical representation of simple mechanical, electrical, and electromechanical systems, transfer function, Block diagram representation and reduction, signal flow graph.	10	3	1

Unit II: Time domain analysis			
Concept of transient response, Steady state and time response, Standard test signals, Type and order of system. Steady state error analysis, Static error constants. Time response of first and second order system. Time response specifications of second order system. Introduction to different types of controllers PD, PI, PID.	9	3	2
Unit III: Stability analysis and Root Locus			
Stability of control systems, condition of stability, characteristic equation, Routh Hurwitz criterion, special cases for determining relative stability. Root location and its effect on time response, elementary idea of root locus, effect of addition of pole and zero on proximity of imaginary axis.	10	3	3
Unit IV: Frequency Domain Analysis			
Concept of frequency response in dynamical systems, Construction of Bode plots, The stability margin on Bode Plot and assessing closed-loop stability from open-loop frequency response, Effect of gain variation and the addition of poles and zeros on frequency response plots (Bode plot). Construction of polar plots and Introduction to Nyquist stability criterion	9	4	4
Unit V: State Variable Analysis			
Concept of state, state variables and state model, Characteristics of system state, Conversion of transfer function to state model using physical variables, Phase variable form and canonical forms, State diagram representation, Relationship between transfer function and state space representation, Transfer function from state space model, Stability analysis of state space model and eigenvalues	7	2	5

Text Books:

1. Modern control system Engineering by K.Ogatta, Publisher - Prentice Hall, India
2. Control System Analysis by Nagrath/ Gopal, Publisher- New age International
3. Automatic Control Systems by B.C.Kuo, Publisher - Prentice Hall, India
4. Control System Engineering by S.K. Bhattacharya, Publisher Pearson

Reference Books:

1. Linear System Design by D'azzo and Houpis, Publisher- McGrawHill
2. Control Systems, Principles & Design by M. Gopal Publisher - TMH(Tata McGrawHill)
3. Control Systems Engineering by Samarajit Ghosh Publisher - Pearson

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit: 1	Practical: 2 Hr		
Subject Code	BEL6P19	Control System Lab	
Examination Scheme			
Practical Internal Marks:	Practical University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	---

Practical Course Objectives	
1	To understand the operation and characteristics of basic control system components such as potentiometer, synchro, and AC servomotor used in feedback control systems.
2	To analyze the time response of control systems and evaluate important parameters such as rise time, settling time, peak overshoot, and steady-state error.
3	To study system stability and performance using techniques such as Routh–Hurwitz criterion and Root Locus method.
4	To analyze the frequency response of control systems using Bode plots and determine stability margins like gain margin and phase margin.
5	To develop the ability to model and analyze control systems using MATLAB, including transfer function representation and state space modeling.

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Identify and analyze the characteristics of basic control system components such as potentiometer, synchro, and AC servomotor.
2	Analyze the time response of first and second order systems using standard test signals.
3	Determine the stability of control systems using techniques such as Routh–Hurwitz criterion and Root Locus method.
4	Evaluate frequency response characteristics of control systems using Bode plot and other frequency domain techniques.
5	Understand and apply state-space modeling techniques to represent dynamic systems, convert between transfer function and state models, and analyze system stability using eigenvalues.

List of Practical:

Pr. No.	List of Practical
1	To study the performance characteristics of a potentiometer as an error detector.
2	To study the performance characteristics of a synchro system.
3	To study the characteristics of a synchro transmitter–receiver pair.
4	To study the synchro transmitter–receiver pair as an error detector.
5	To determine the speed–torque characteristics of a small AC servomotor.
6	To obtain and analyze the transient response of a first order system using MATLAB.

7	To plot the transient response of a second order system using MATLAB and determine rise time, settling time and peak overshoot.
8	To obtain standard test signal responses (step, ramp, impulse) of a given transfer function using MATLAB.
9	To determine the steady state error and static error constants (K_p , K_v , K_a) of a given control system.
10	To plot the Root Locus of a given transfer function using MATLAB and study the effect of pole and zero locations.
11	To plot the Bode plot of a given transfer function and determine gain margin and phase margin.
12	To verify the stability of a control system using Routh–Hurwitz Criterion.
13	To obtain the transfer function from a given state space model using MATLAB.
14	To obtain the state space model from a given transfer function using MATLAB.
15	To study the effect of PID controller on system performance using MATLAB.

* Any Ten practical's from the above-mentioned practical's can be performed

Text Books

1. Modern Control Engineering by K. Ogata, Publisher – Prentice Hall of India.
2. Control Systems Engineering by N. S. Nise, Publisher – John Wiley & Sons.
3. Automatic Control Systems by B. C. Kuo and Farid Golnaraghi, Publisher – John Wiley & Sons.
4. Control Systems by I. J. Nagrath and M. Gopal, Publisher – New Age International.

Reference Books:

1. Feedback Control of Dynamic Systems by G. F. Franklin, J. D. Powell and A. Emami-Naeini, Publisher – Pearson Education.
2. Modern Control Systems by Richard C. Dorf and Robert H. Bishop, Publisher – Pearson Education.
3. Control System Engineering by N. K. Sinha, Publisher – New Age International.

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FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit :3	Lecture (L): 3Hr	Practical /Activity (T/A): 0	
Subject Code	BEL6T20A	Electrical Safety & Standards	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	---

Course Objectives

1	To make the students to realize the importance of electrical safety and standards.
2	To make the students to understand the safety aspects associated with the electric shock, fire and other accidents at various installations.
3	To introduce the students with the electrical safety measures to be taken during Installation, Testing, Commissioning, Operation and Maintenance
4	To make the students aware about the hazards and safety precautions to be taken at electrical installations.
5	Familiarize students with Safety Management policies, Indian Electricity (IE) Rules, and mandatory earthing practices.

Course Outcomes	
After completion of syllabus, students would be able to	
1	Understand the principles of safety, Shocks and their prevention
2	Understand the safety aspects associated with Residential, Commercial and Agricultural Installations
3	Outline the electrical safety during installation, testing, commissioning, operation and maintenance.
4	Categorize hazardous zones and select appropriate equipment enclosures
5	Understand Safety Management policies, Indian Electricity (IE) Rules & acts, and earthing practices.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Introduction To Electrical Safety, Shocks And Their Prevention:			

Terms and definitions, objectives of safety and security measures, Hazards associated with electric current and voltage, protection against Electrical Hazards, , principles of electrical safety, Approaches to prevent Accidents, scope of subject electrical safety. Primary and secondary electrical shocks, possibilities of getting electrical shock and its severity, Effects of electric current on the human body, medical analysis of electric shocks and its effects, First principles of actions after electric shock, shocks due to flash/ Spark over's, prevention of shocks, safety precautions against contact shocks, flash shocks, burns.	9		1
Unit II: Electrical Safety in Residential, Commercial and Agricultural Installations:			
Introduction, Wiring and fitting , Domestic appliances, water tap giving shock, shock from wet wall, fan and cooler giving shock, multi-storied building, Temporary installations, Agricultural pump installation, Do's and Don'ts for safety in the use of domestic appliances.	9		2
Unit III: Electrical Safety during Installation, Testing and Commissioning, Operation and Maintenance:			
Preliminary preparations, safe sequence, risk during installation of plant and equipment –safety documentation –field quality and safety during erection, personal protective equipment for erection personnel, safety clearance notice, safety precautions during plant energizing, safeguards for operators safety.	9		3
Unit IV: Electrical Safety in Hazardous Areas:			
Hazardous zones, class 0,1 and 2 , spark, flashovers and corona discharge and functional requirements, Specifications of electrical plants, equipment's for hazardous locations, Classification of equipment enclosure for various hazardous gases and vapours, classification of equipment/enclosure for hazardous locations.	9		4
Unit V: Safety Management and Standards of Electrical Systems :			
Principles of Safety Management, Management Safety Policy, Safety organization, safety auditing, Motivation to managers, supervisors, employees. Introduction to Indian Electricity Rules, Acts and their Significance, Objectives & Scope of Indian Standards (IS) developed by BIS, Scope & applications of International Standard IEC 61439-1 and IEC 61439-2 , Significance of equipment earthing and earthing of non-current carrying metallic parts, Rules regarding first aid and fire fighting facility.	9		5

Suggested Text books & Reference Books:

Text Books

- 1.“ Electrical Safety & Standards” Dr.R. Keswani , Prof.K.Gaurkhede , Dr.U.Hiwase , Sai Jyoti Publication , August 2024
2. “Electrical Safety, Fire Safety Engineering and Safety Management”, S. Rao, R. K. Jain, and H. L. Saluja, Khanna Publishers
3. “Energy management policy, planning and utilization”, Pradeep Chaturvedi, Concept Publishing company, New Delhi.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit :3	Lecture (L): 3Hr	Practical /Activity (T/A): 0	
Subject Code	BEL6T20B	Electric Drives & Control	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3Hr

Course Objectives

1	To understand the concept of Electrical drives and drives commonly used in industries.
2	To relate various electrical and mechanical characteristics of electrical drives.
3	To analyze motor control used in Electric Vehicle and industrial automation by using PLC.
4	To Acquire knowledge of digital control of electrical drives used in industries along with use of AC / DC contactors in electric drives.
5	To perceive the motors used in Electric Traction and its control strategies.

Course Outcomes

After completion of syllabus, students would be able to

1	Understand the concept of Electrical drives and drives commonly used in industries.
2	Relate various electrical and mechanical characteristics of electrical drives.
3	Analyze the causes and effects of motor control used in Electric Vehicle and industrial automation by using PLC.
4	Acquire knowledge of digital control of electrical drives used in industries along with use of AC / DC contactors in electric drives.
5	Perceive the concept of Electric traction and their control strategies used in practice.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Fundamentals of Electrical Drives			
Introduction, Classification, Advantages, Disadvantages, Types of motors used, Parts of electrical drives, Status of ac and dc drives, Types of loads / load torques,	9	0	1

Transmission of Drive, Energy conservation in electrical drives. Brief Idea about drives Commonly used in industries.			
Unit II: Characteristics of Electrical Drives			
Speed- torque characteristics of common drive motors and their characteristics under Starting, Running, Braking and Speed control. Selection of Motor, Load Equalization, Flywheel Effects and Flywheel Calculations	9	0	2
Unit -III : Introduction to Electric Vehicle and Traction Drives			
Block Diagram, Components of EV, Classification, Electric Motors used, Speed Control, Controllers & Battery charging. Different system of traction, Types of railway services, Speed-time curves for train movement, Traction motor control, Series parallel control with numerical, starting and braking of traction motor.	9	0	3
Unit IV: AC / DC contactors			
Lock out contactors, magnetic structure, operation, arc interruption, contactorrating, HV contactors, Control circuit for automatic starting and braking of DC motor and threephase induction motor drives.	9	0	4
Unit V: PLC and Digital Control			
Introduction, Basic Block diagram, Programming, application and advantages in electrical drives. Digital control of electric drives. Block diagram, comparison with other methods of control.	9	0	5

Text Books :

1. Dubey G.K, “Fundamentals of Electrical Drives”, Narosa Publishing House
2. Pillai S.K., “A First Course on Electrical Drives”, New Age International
3. De N.K., Sen P.K. “Electric Drives”, Prentice Hall of India
4. H.Partab, “Modern Electrical Traction”,Pritam Surat & Brothers.
5. M.L. Soni, P.V. Gupta, U.S.Bhatnagar, “A course in Electrical Power”, Dhanpat Rai & Sons.
6. K Wang Hee Nam: AC Motor Control & Electrical Vehicle Application, CR Press, Taylor & Francis Group.

Reference Book :

- 1.Vedam Subrahmanyam, “Electric Drives – Concepts & Applications”, Tata McGraw-Hill.
2. H.Partab, “Art & Science of Utilization of Electrical Energy”, Dhanpat Rai & Sons.
3. Krishnan. R, “Electric Motor Drives: Modeling, Analysis and Control”, Prentice Hall of India.
4. Ward Leonard, “Control of electrical drives”, Springer.
- 5 Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press.
6. James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley.
7. C.C Chan, K.T Chau: Modern Electric Vehicle Technology, Oxford University Press Inc., New York.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit : 3	Lecture (L): 3 Hr	Practical /Activity (T/A): 0	
Subject Code	BEL6T20C	Energy Management & Audit	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hr

Course Objectives

1	To understand energy demand, supply and the importance of energy conservation.
2	To learn principles and methodology of energy audit in electrical and thermal systems.
3	To develop ability to analyze energy performance and energy efficiency opportunities.
4	To understand modern energy management technologies such as smart grids and renewable integration.
5	To understand energy demand, supply and the importance of energy conservation.

Course Outcomes

After completion of syllabus, students would be able to

1	CO1: Explain energy demand–supply scenario and the need for energy conservation.
2	CO2: Apply different types of energy audit techniques to analyze industrial energy usage.
3	CO3: Prepare material and energy balance diagrams for energy systems.
4	CO4: Develop energy action plans and monitoring strategies for industries and buildings.
5	CO5: Evaluate energy efficiency technologies and emerging smart energy systems.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I: Basics of Energy Management and Conservation			
Global and Indian energy scenario, Energy demand and supply trends, Environmental issues and climate change, Concept of energy management,	9	0	1

Energy conservation principles, Energy consumption in residential and commercial buildings can be optimized using smart monitoring systems.			
Unit II: Energy Audit and Energy Performance Assessment			
Definition and importance of energy audit, Types of energy audit – Preliminary audit and Detailed audit , Energy audit methodology, Benchmarking techniques Energy Conservation Act (India) and role of Bureau of Energy Efficiency (BEE) Modern energy audits: IoT based energy monitoring systems, Smart sensors for real time energy auditing, Digital energy audit tools.	9	0	2
Unit III: Material and Energy Balance			
Concept of material balance and Energy balance principles in industry , Preparation of energy flow diagrams using industrial data, material balance diagrams and energy balance diagrams with conventional energy generation, Cogeneration and Waste heat recovery in Power plant and industries. Energy balance studies in renewable generation and hybrid energy systems.	9	0	3
Unit IV: Energy Action Planning, Monitoring and Targeting			
Energy management policy, Energy planning process, Force field analysis, Development of energy monitoring dashboard, Industry Application: Monitoring energy consumption in smart industries, Implementation of ISO 50001 Energy Management System, Develop energy action plans and monitoring strategies. Smart grid technology enables efficient monitoring and control of energy consumption.	9	0	4
Unit V: Electrical and Thermal Energy Management with Emerging Technologies			
Electrical Energy Management: Power factor improvement, Energy efficient electric motors ,Variable frequency drives, Load management Thermal Energy Management: Energy conservation in boilers, steam turbines and furnaces. Role of Energy Auditor & Energy Manager	9	0	5

Text Books

1. Amit Kumar Tyagi, Handbook on Energy Audits and Management, TERI Publication.
2. Wayne C. Turner, Energy Management Handbook, Wiley Publication.

Reference Books

1. Bureau of Energy Efficiency (BEE) – Energy Manager Training Material.
2. Archie W. Culp – Principles of Energy Conservation.
3. P. O’Callaghan – Energy Management.
4. Thuman A. & Mehta D. Paul – Handbook of Energy Engineering.
5. Y.P. Abbi & Shashank Jain – Handbook of Energy Audit and Environment Management.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit : 4	Lecture (L): 3 Hr	Practical /Activity (T/A): 1	
Subject Code	BEL6T21A	Electric Vehicles	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	---

Course Objectives

1	To compare and analyze different types of EVs.
2	To evaluate the impact of various parameters on the performance of EVs.
3	Investigate various components used in EV's
4	Understand energy storage system in electric vehicle
5	Understanding the function and requirement of converter in EVs

Course Outcomes

After completion of syllabus, students would be able to

1	Measure economic and environmental impact of EVs compared to IC engines
2	Analyze the mathematical models of EVs and justify impact of various parameters
3	Compare various types of EV propulsion drive motors and choose the best for given application.
4	Compare various types of EV storage systems and analyze characteristics
5	Apply various Power electronic circuit topologies for EVs

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit-I : Introduction to Electric Vehicle (EV):			
History of EVs – Economic and Environmental impact; Architecture of EV. Basics of vehicle performance, vehicle power source characterization, transmission characteristics. Electric Drivetrains- Basic concept of electric traction, introduction to various electric drive-train topologies	10	4	1

Unit-2 :Electric Vehicle Modelling:			
Consideration of Rolling Resistance .Transmission Efficiency. Consideration of Vehicle Mass , Tractive Effort. Modelling of Vehicle Acceleration . Modelling of Electric Vehicle Range , Aerodynamic Considerations . EV Motor Sizing.	9	3	2
Unit-3 :Electric Propulsion:			
Introduction to electric components used in electric vehicles, Configuration and control of DC Motor drives, Configuration and control of Induction Motor drives, configuration and control of Permanent Magnet Motor drives, Configuration and control of Switch Reluctance Motor drives	9	3	3
Unit-4 : Energy Storage:			
Introduction to Energy Storage Requirements in Electric Vehicles, Battery based energy storage and its analysis, Super Capacitor based energy storage and its analysis, Flywheel based energy storage and its analysis, Hybridization of different energy storage devices.	9	3	4
Unit-5 : Power Converters:			
DC-DC converters for EVs, Buck and Buck-Boost Converters, Multi-quadrant DC-DC converters, DC-DC converter applications, DC-AC converters for EVs, Three-phase DC-AC converters, Voltage control of DC-AC inverters using PWM.	8	2	5

Text Books:

1. Iqbal Hussein, “Electric and Hybrid Vehicles: Design Fundamentals”, CRC Press, 2nd Edition, 2003.
2. James Larminie, John Lowry, “Electric Vehicle Technology”, Wiley publications, 1st Edition, 2003.
3. Dr. Nitin Dhote et.al., “ Electrical Vehicle Technology” , R.K.Publications,1st Edition, 2025.
4. Praful Nandankar, “Application of Power Electronics for Renewable Energy Systems”, Notion Press, 1st Edition, 2022.
5. Keerthi Kumar N et. al., “E - Vehicle Renewable Energy”, Scientific International Publishing House, 1st Edition, 2024.

Reference Books:

1. B D McNicol, D A J Rand, “Power Sources for Electric Vehicles”, Elsevier publications, 1st Edition, 1998.
2. Seth Leitman, “Build Your Own Electric Vehicle” MC Graw Hill, 1st Edition, 2013.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit : 4	Lecture (L): 3 Hr	Tutorial (T): 1 Hr	
Subject Code	BEL6T21B	HV Engineering	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3Hrs

Course Objectives

1	To understand the breakdown mechanism in Di-electric medium
2	To study lighting and switching over voltages
3	To study generation of high voltage and currents
4	To study measurement of high voltage and current
5	To study Insulation Coordination and high voltage testing of electrical apparatus

Course Outcomes

After completion of syllabus, students would be able to

1	Understand breakdown mechanism in gases Liquid and Solid medium.
2	Know of lightening and switching over-voltages.
3	Analyse different methods of generation of high voltage and currents in laboratory.
4	Apply different methods of measurement of high voltage and currents in laboratory.
5	Analyse different methods of Insulation Coordination and High Voltage testing of Apparatus and cables in laboratory Contents.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T	CO
Unit I : Breakdown mechanism in Di-electric			
Electrical insulation and dielectrics, Properties of dielectric materials, Uniform and non-uniform Electric field, ionization Process (Townsend, Streamer theory) Vacuum dielectric Break down mechanism,	9	3	1

Liquid: Types of liquid dielectrics ,Breakdown as mechanisms, Factors affecting dielectric strength of liquids (Transformer oil). Solid: Breakdown mechanism, breakdown of solid dielectric in practice, Breakdown due to treeing & tracking, breakdown due to the internal discharge.			
Unit II: Lighting and switching over voltages:			
Mechanism of lightening, types of strokes, parameter and characteristics of lightening strokes, characteristics of switching surges; power frequency over-voltages. Control of O.V. due to switching. Protection of lines by ground wires, protection by lightning Arrester, Types of L.A., selection of L.A. ratings, surge-absorbers. Volt-time Characteristics.	9	3	2
Unit III: Generation of high voltage and. Currents:			
Generation of High D.C voltages by rectifiers, voltage doubler and multiplier circuits (Derivations and expression not required) Generation of high AC voltages by Cascade transformers, Resonant transformers, Tesla Coil. Generation of high impulse voltages: Single and multistage, Marx circuit, generation of switching surges generation of impulse current.	9	3	3
Unit IV: -Measurement of high voltage and current:			
Measurement of high AC and DC voltage by micro ammeter, generating voltmeter resistance and capacitance potential divider, CVT. Peak reading AC voltmeter. Sphere gap arrangement. Measurement of impulse voltage by potential dividers and peak reading voltmeters. Measurement of High AC DC current: measurement of high frequency and impulse current.	9	3	4
Unit V: High Voltage testing and Insulation Coordination :			
High voltage testing of electrical apparatus as per International & Indian standards Power frequency tests, Impulse voltage tests. DC tests on: insulators, circuit breakers, bushings, isolators, transformers Insulation coordination: basic insulation levels, surge protection & coordination methods.	9	3	5

Suggested Text books & Reference Books

Text Books

1. High Voltage Engineering, M.S. Naidu and V Kamaraju, TMG
2. High Voltage Engineering, C.L.Wadhwa New Age International

Reference Books

1. Advances In high Voltage Engineering A. Haddat and D. Warne, IET

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit : 4	Lecture (L): 3 Hr	Practical /Activity (T/A): 1	
Subject Code	BEL6T21C	Electrical Distribution Systems	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3Hrs

Course Objectives

1	Students will be able to learn various aspects of distribution system
2	Students will be able to distribution system analysis
3	Students will be able to understand the modeling of various distribution system components
4	Students will be able to understand the various methods
5	Students will be able to learn current trends in distribution system

Course Outcomes	
After completion of syllabus, students would be able to	
1	Discuss the fundamental components and configurations of distribution systems, enabling the classification of various loads
2	Apply analytical methods to calculate series impedance, determine feeder loading, evaluate K-factors for non-linear loads, and perform load flow studies to optimize distribution system performance.
3	Model single and three-phase distribution transformers, voltage regulators, and capacitor banks to determine their impact on power distribution performance.
4	Analyze power factor improvement and voltage control methods to enhance system efficiency and performance.
5	Discuss the smart energy systems strategies .

SYLLABUS

Details of Topic	Allotment of Hours	Mapped with CO Number	
	L	T/A	CO
Unit I: Introduction to Electrical distribution system			

Introduction to Electrical distribution system, Components of distribution system substation and busbar layouts, Components of distribution system and feeder configurations, Relationship between the load factor and loss factor - Classification of loads, Changes in load curve due to loads.	9	3	1
Unit II: System Analysis			
Computation of transformers and feeder loading, ‘K’ Factors and Their Applications, Analysis of Uniformly Distributed Loads, Distribution of loads and various geometric configurations, Series Impedance of Distribution Lines and Feeders, Series Impedance of Distribution Lines and Feeders, Load flow analysis.	9	3	2
Unit III: System Modeling			
Modeling of distribution system components such as Overhead lines, feeders and cable, Single and three phase distribution transformer, Voltage regulator, Load models, Capacitor banks	9	3	3
Unit IV: Power factor Improvement & Voltage Control methods			
Causes and effects of low power factor, most economical power factor, Methods of improving power factor:-static capacitor, synchronous condenser, phase advancer; Voltage control methods;-Excitation control, shunt capacitor and reactors, tap changing transformers, booster transformer, synchronous condensers	9	3	4
Unit V: Recent trends in distribution system			
Rooftop Solar Storage: Managing bidirectional power flows. Microgrids: Development of local, controllable, and resilient energy networks. Virtual Power Plants (VPPs), Load shedding and management	9	3	5

Suggested Text books & Reference Books

Text Books

1. W. H. Kresting, Distribution System Modeling and Analysis, CRC Press, New York, 2002.
2. Electrical Power Distribution System Kamaraju. Tata-McGraw Hill Publications
3. Electric Power Distribution A. S. Pabla, Tata Mc Graw-Hill Publishing
4. A. A. Sallam and O. P. Malik, Electric Distribution System, IEEE Press, Piscataway, NJ, 2011.
5. J. H. Teng, “A direct approach for distribution system load flow solutions,” IEEE Trans. on Power Delivery, vol. 18, no. 3, pp. 882–887, 2003.
6. R. F. Arritt and R. C. Dugan, “Distribution system analysis and the future smart grid,” IEEE Trans. on Industry Applications, vol. 47, no. 6, pp. 2343-2350, November/ December 2011.

RASHTRASANT TUKADOJI MAHARAJ NAGPUR UNIVERSITY, NAGPUR
FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit: 2	Lecture (L): 2 Hr	Practical /Activity (T/A): 0	
Subject Code	BMD6T22	Microprocessor and Microcontroller	
Examination Scheme			
Internal Marks:	University Marks:	Minimum Passing Marks:	Examination Duration:
30 Marks	70 Marks	45 Marks	3 Hrs

Course Objectives

1	Understand the architecture and instruction set of the 8085 microprocessor.
2	Explain interrupts and peripheral interfacing using 8085 and 8255.
3	Understand the architecture and programming model of the 8051 microcontroller.
4	Develop interfacing and basic embedded programming using 8051 and Arduino.

Course Outcomes

After completion of syllabus, students would be able to

1	Explain the architecture and programming of the 8085 microprocessor.
2	Implement interrupts and I/O interfacing using 8085 and 8255.
3	Analyze the architecture and programming features of the 8051 microcontroller.
4	Interface peripheral devices and develop basic embedded applications using 8051 and Arduino.

SYLLABUS

Details of Topic	Allotment of Hours		Mapped with CO Number
	L	T/A	CO
Unit I			
Introduction to Intel's 8085A Architecture description, Flag register, addressing modes, complete instruction set, Instruction cycle and timing diagram, Stack operation with PUSH & POP instruction, CALL/RETURN instruction, Assembly Language programming of 8085.	8	0	1
Unit II			

8085 Interrupts and ISR, Memory interfacing with 8085, 8255 PPI architecture, Pin descriptions, operating modes, BSR and I/O modes, IN and OUT instructions, Interfacing of 8255 with 8085, Interfacing of LEDs using 8255 PPI.	7	0	2
Unit III			
Difference between microprocessor & microcontroller, Architecture, PSW, Instruction Set, Addressing modes and Assembly Language Programming, Stack, Interrupt, Timers, Serial Communication, SFRs.	8	0	3
Unit IV			
Interfacing and Programming of LED, Stepper & DC Motor, A/D and D/A converters with 8051. Introduction to Arduino programming for basic control applications.	7	0	4

Suggested Textbooks & Reference Books

Text Books

1. Programming and interfacing 8085A, Gaonkar, Wiley Eastern
2. Programming of 8085, D.V. Hall, McGraw-Hill
3. Microprocessor Principles and Applications Pal Tata McGraw-Hill
4. The 8051 Microcontroller and Embedded System, M.A. Mazidi& J.G. Mazidi, Pearson Education

Reference Books

1. Intel Reference Manuals, Microprocessors & Microcontrollers: Intel
2. Microcontrollers – Peatman, McGraw-Hill.
3. Microprocessors & Microcomputers-based system design by Md. Rafiquzzaman.
4. The 8051 Microcontroller & Embedded Systems, Kenneth J. Ayala, Dhanvijay V. Gadre, CENGAGE Learning
5. Microprocessors Principles and Applications Gomorra Tata McGraw-Hill

FACULTY OF SCIENCE & TECHNOLOGY
B.TECH. ELECTRICAL ENGINEERING

Sem: VI	Total Hours Distribution per week		
Total Credit: 2	Practical: 4 Hr		
Subject Code	BSE6P02	Electrical Installation Design lab	
Examination Scheme			
Practical Internal Marks:	Practical University Marks:	Minimum Passing Marks:	Examination Duration:
25 Marks	25 Marks	25 Marks	---

Practical Course Objectives	
1	To develop the ability to analyze electrical loads and prepare load schedules for residential
2	To enable students to select appropriate cables
3	To impart skills in designing electrical systems including busbars
4	To provide hands-on experience in designing complete electrical installations
5	To familiarize students with modern software tools such as AutoCAD Electrical

Practical Course Outcomes	
After completing the practical course, students will be able to	
1	Calculate connected load, maximum demand, diversity factor, and prepare load schedules for various installations
2	Select suitable cables, protective devices, and design protection schemes for safe and efficient operation
3	Design power and control circuits for DOL and star-delta starters, and perform power factor correction.
4	Design electrical layouts including substations, earthing systems, and complete small industrial installations with cost estimation.
5	Use software tools (AutoCAD Electrical, ETAP, MATLAB/Simulink) to develop SLDs, perform load flow studies, and simulate electrical systems.

A minimum of eight experiments shall be conducted, subject to the availability of laboratory equipment and relevant software resources. The following experiments are recommended:

List of Practical:

Pr. No.	List of Practical
1	To determine the connected load, maximum demand, and diversity factor for a given installation.
2	To prepare a load schedule for a residential/commercial/industrial installation.
3	To select the appropriate size of cable for a given load based on current carrying capacity. (Electrical Machine lab/ college/ commercial complex/ small scale industry)
4	To design a busbar system for a given industrial load.

5	To study and compare different methods of cable laying for electrical installations. To select suitable switching and protective devices (MCB/MCCB/Fuse) for a given installation.
6	To design a protection scheme against overload, short circuit, and earth fault.
7	To design the power and control circuit for a Direct-On-Line (DOL) starter.
8	To design the power and control circuit for a star-delta starter.
9	To determine the required kVAR rating of a capacitor bank for power factor improvement.
10	To design a complete electrical installation for a small industrial unit.
11	To prepare the layout of an 11 kV / 33 kV substation.
12	To design an earthing system for a substation.
13	To design and prepare a detailed electrical installation plan for a small industry including load calculation, protection, and cost estimation.
14	To familiarize with BIS standards for electrical installation practices.
Software based	
1	To develop a single line diagram (SLD) of an industrial electrical system using AutoCAD Electrical.
2	To perform load flow analysis for a small power system using ETAP / MATLAB.
3	To simulate protection coordination of relays and circuit breakers using ETAP / MATLAB.
4	To analyze power factor improvement and capacitor bank sizing using ETAP / MATLAB.
5	To design a substation layout using AutoCAD.
6	To simulate the performance of an induction motor and its starting methods using MATLAB / Simulink